

B.Sc. Mathematics – 2nd Semester

MTB 202 – Statics and Dynamics

by

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Part – I

Forces in Two-Dimension



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Equilibrium of forces acting at a point

Analytical conditions: If a number of forces act at a point then, they are in equilibrium if and only if algebraic sums of their resolved parts along two perpendicular directions through the point are zero.

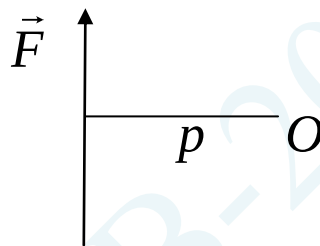
$$R = \sqrt{R_x^2 + R_y^2}, R_x = 0, R_y = 0.$$

Geometrical conditions: If any number of forces acting at a point can be represented in magnitude and direction by the sides of a polygon, taken in order, then the forces will be in equilibrium.



Moment of a force about a point

The moment of a force about a point is defined as the product of the force and the perpendicular distance of the point from the line of action of the force.

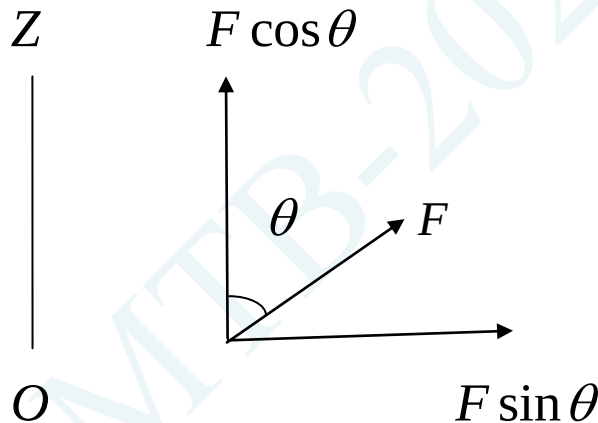


If $\vec{F}$ be the force and O be the point, then $F.p$ is the moment of the force about O , where p is the perpendicular distance of O from $\vec{F}$. [$|\vec{F}| = F$]



Moment of force about a line

Let $\vec{F}$ be any force and OZ be any line (in space). Let θ be the angle between them. Let us resolve $\vec{F}$ into $F \cos \theta$ parallel to OZ and $F \sin \theta$ perpendicular to OZ . [$|\vec{F}| = F$]



Let p be the shortest distance between the line of action of $\vec{F}$ and OZ . Then $F \sin \theta.p$ is called the moment of F about OZ .

“The moment of a force about a line is the product of the resolve part of the force perpendicular to the line and the shortest distance between the line of action of the force and the line”.

It is evident that if the line of action of the force intersects the line about which the moment is considered, then the shortest distance between them is zero. Hence the moment of the force about the line is equal to zero $[F \sin \theta.0 = 0]$.



On the other hand, if the force is parallel to the line then $\theta = 0$ and hence the moment of the force about the line is equal to zero $[F \sin 0 \cdot p = 0]$.

Theorem: If two forces act at a point, the algebraic sum of their moments about any line in space is equal to the moment of their resultant about the same line.

Note: This theorem holds for any number of concurrent forces.



Couples

Two equal unlike parallel forces whose lines of action are not the same, form a Couple.

A Couple is called a torque by some writers; by others, the word torque is used to denote the moment of a Couple.

Examples

- (i) The forces by the hands to the handle of a car while driving,
- (ii) The forces by the fingers to the key during locking,
- etc.



The arm of a couple is the perpendicular distance between the lines of action of the two forces which form the couple.

The moment of a couple is the product of one of the forces forming the couple and the arm of the couple.

If the force and the arm of a couple are P and p respectively, then the couple is denoted by (P, p) and its moment is $P.p$.

Couples acting in the same plane and having equal moments are equivalent. Consequently, couples of equal but opposite moments in the same plane are balance each other.



Resultant of Couples (Coplaner)

Let us consider the couples (P, p) , (Q, q) , (R, r) , $(S, s), \dots$, where $P, Q, R, S, \dots$ are the forces of the couples and $p, q, r, s, \dots$ are their arms. [As the couple of equal moments in a plane are equivalent]. Now, replace the couple (Q, q) by a couple whose components have the same lines of action as those of (P, p) . If X be the magnitude of the force of this couple, then we must have

$$X.p = Q.q$$

$$\text{So that, } X = \frac{Qq}{p}.$$



Similarly, replace the couples (R, r) , $(S, s), \dots$ by couples whose forces act in the same line as that of (P, p) . We are then left with a single couple the magnitude of whose force is

$$P + Q \cdot \frac{q}{p} + R \cdot \frac{r}{p} + S \cdot \frac{s}{p} + \dots \text{ and the moment of the couple is}$$

$$\left(P + Q \cdot \frac{q}{p} + R \cdot \frac{r}{p} + S \cdot \frac{s}{p} + \dots \right) p = P \cdot p + Q \cdot q + R \cdot r + S \cdot s + \dots$$

Theorem: The resultant of any number of couples acting in the same plane on a rigid body is a couple whose moment is equal to the algebraic sum of the moments of the couples.



To give a couple in a plane it is not necessary to give the positions or magnitude of the forces of the couple but only the moment of the couple should be given.

Again, couples acting in parallel planes and have equal moments are equivalent. So, it is not even necessary to state the plane of the couples it is enough to give the direction of the normal to its plane.

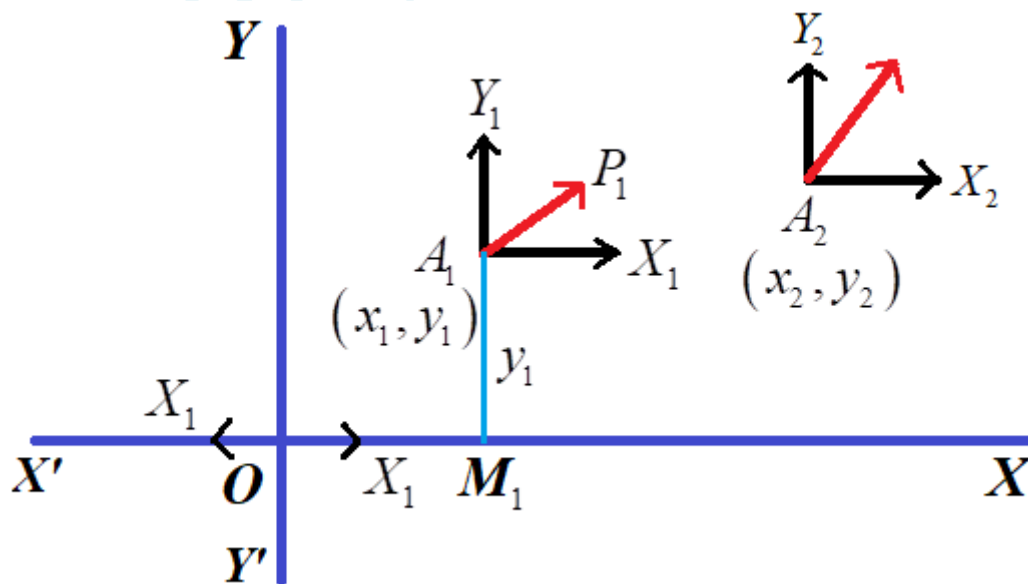
Thus, a couple is completely determined by the direction of the normal to its plane of action and the moment with its sense.



Reduction of Coplanar Forces

Theorem: A system of forces acting in one plane at different points of a rigid body can be reduced to a single force through a given point and couple.

Proof: Let the forces $P_1, P_2, P_3, \dots$ act at points $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots$ of the body, the coordinates of the points being given



with reference to rectangular axes OX and OY through a given point O .

Consider first the force P_1 acting at $A_1(x_1, y_1)$. Let it be resolved into two forces X_1 and Y_1 parallel to the coordinate axes. At O introduce two equal and opposite forces X_1 , one along OX and other along OX' .

This will have no effect on the body. Now the forces X_1 at A_1 and X_1 at O along OX' form a couple of moment $-X_1y_1$ and we are left with the force X_1 along OX .

Thus the force X_1 at A_1 is equivalent to a force X_1 at O along OX and a couple of moment $-X_1y_1$.



Similarly by introducing at O equal forces Y_1 along OY and OY' , it is easy to see that the force Y_1 at A_1 is equivalent to a force Y_1 at O along OY and a couple of moment $Y_1 x_1$.

It follows, therefore, that the force P_1 at A_1 is equivalent to force X_1 and Y_1 at O along OX and OY respectively and a couple of moment $Y_1 x_1 - X_1 y_1$.



Proceeding in the same way with the remaining forces we see that the given system of forces is equivalent to

$$R_x = X_1 + X_2 + X_3 + \dots = \Sigma X_i \quad \text{along } OX$$

$$\text{and } R_y = Y_1 + Y_2 + Y_3 + \dots = \Sigma Y_i \quad \text{along } OY,$$

and a couple of moment

$$G = (Y_1x_1 - X_1y_1) + (Y_2x_2 - X_2y_2) + \dots = \Sigma (Y_ix_i - X_iy_i) \quad (1)$$

The forces R_x and R_y can be compounded into a single force through O of magnitude R given by

$$R^2 = R_x^2 + R_y^2 \quad (2)$$



acting at an angle $\theta = \tan^{-1} \left(\frac{R_y}{R_x} \right)$ with the X -axis.

Thus the system of forces can be reduced to a single force R through O and a couple of moment G .

It is evident that G depends upon the position O of the given point while R does not.



Theorem: A system of forces acting in one plane at different points of a rigid body can be reduced to a single force, or a couple.

Proof: We have already seen that a system of coplanar forces acting in a body in general can be reduced to a single force R and a couple of moment G , given by equation (2) and (1).

If $R = 0$. Then the forces reduce to a couple and the theorem is proved. But if $R \neq 0$, we shall show that the force R and the couple of moment G can be reduced to a single force R acting in the same direction but in a different line.

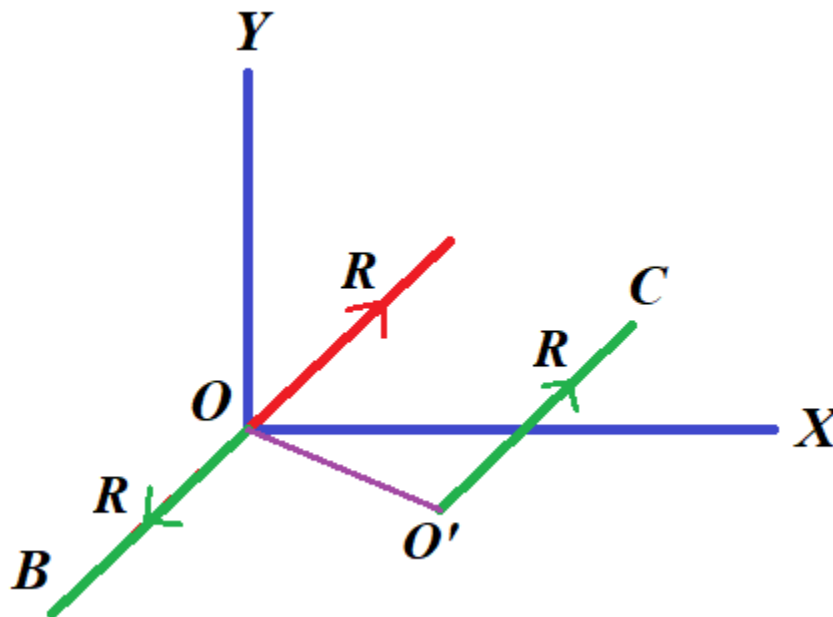


Replace the couple G by two equal and opposite forces of magnitude R , one along OB in the direction opposite to R and the other along $O'C$, where OO' is the perpendicular to OB i.e., the arm of the couple.

$$\text{Then } OO' = \frac{G}{R}$$

$$\text{i.e. } OO' = \frac{\Sigma(Y_1x_1 - X_1y_1)}{\sqrt{R_x^2 + R_y^2}}$$

The forces at O balance each other and we are left with the force R acting at O' along $O'C$.



[The equation of the line $O'C$ may be obtained as follows

Since $\tan \theta = \frac{R_y}{R_x}$, we have $\sin \theta = \frac{R_y}{R}$ and $\cos \theta = \frac{R_x}{R}$.

The line $O'C$ is given by

$$x \frac{R_y}{R} - y \frac{R_x}{R} = \frac{G}{R} \Rightarrow xR_y - yR_x = G.]$$

Equation of the resultant

Now we obtain the equation of the final resultant of a system of coplanar forces.



Let $P_1, P_2, P_3, \dots$ be the forces acting at points $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots$ of a body.

Let G be the moment of all the forces about the origin.

$$\text{So } G = \Sigma(Y_1 x_1 - X_1 y_1) \quad [\text{similar as previous theorem}]$$

If the coordinates of a point Q be given by (ξ, η) the moment of force P_1 about Q is

$$Y_1(x_1 - \xi) - X_1(y_1 - \eta)$$

$$\text{i.e., } (Y_1 x_1 - X_1 y_1) - Y_1 \xi + X_1 \eta$$



Writing the moment of the other forces similarly, we see that the total moment G' of all the forces about the point Q is given by

$$G' = G - \xi \Sigma Y_1 + \eta \Sigma X_1 = G - \xi R_y + \eta R_x.$$

If the point Q lies on the resultant, then we have $G' = 0$.

$$\text{Hence, } G - \xi R_y + \eta R_x = 0.$$

Replacing ξ, η by x, y we can see that the equation line of action is same as found before.



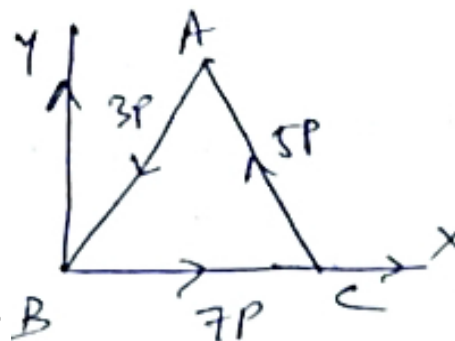
Example: Forces equal to $3P$, $7P$ and $5P$ act along the sides AB , BC and CA of an equilateral triangle ABC , find the magnitude and direction of the resultant.

Solution: Let O be the centroid of the equilateral. Let us consider the rectangular axes at B , X -axis along BC and Y -axis perpendicular to BC .

$$\text{Then } R_x = 7P \cos 0^\circ + 5P \cos 120^\circ + 3P \cos 240^\circ$$

$$= 7P - \frac{5}{2}P - \frac{3}{2}P = 3P$$

$$R_y = 0 + 5P \sin 120^\circ + 3P \sin 240^\circ = \frac{5\sqrt{3}}{2}P - \frac{3\sqrt{3}}{2}P = \sqrt{3}P$$



Therefore the magnitude of the resultant is

$$R = \sqrt{(3P)^2 + (\sqrt{3}P)^2} = \sqrt{12}P$$

and it makes an angle $\tan^{-1}\left(\frac{\sqrt{3}P}{3P}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30^\circ$ with BC .



Example: If six forces of relative magnitude 1, 2, 3, 4, 5 and 6 act along the sides of a regular hexagon, taken in order, show that the single equivalent force is F relative magnitude 6, and that it acts along a line parallel to the force 5 at a distance from the centre of the hexagon $\frac{7}{2}$ times the distance of the sides from the centre.

Solution: Let $ABCDEF$ be a regular hexagon of side $2a$. Suppose that forces of magnitude $k, 2k, 3k, 4k, 5k$ and $6k$ act along the sides AB, BC, CD, DE, EF and FA respectively.

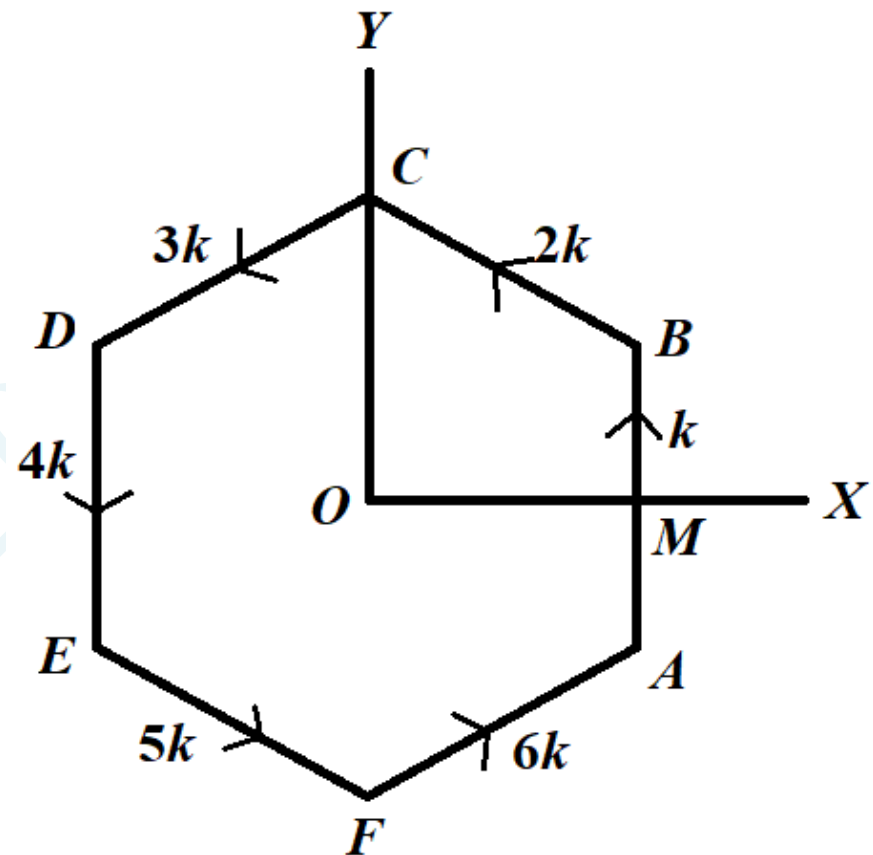


Let O be the centre of the hexagon. Choose through O a set of rectangular axes OMX and OCY .

It is easy to see that the six forces are inclined to OMX at the angles

$\frac{1}{2}\pi, \frac{5}{6}\pi, \frac{7}{6}\pi, \frac{9}{6}\pi, \frac{11}{6}\pi$ and $\frac{13}{6}\pi$.

Hence the sums of resolved parts along the axes are



$$R_x = k \left(\cos \frac{\pi}{2} + 2 \cos \frac{5}{6} \pi + 3 \cos \frac{7}{6} \pi + 4 \cos \frac{9}{6} \pi + 5 \cos \frac{11}{6} \pi + 6 \cos \frac{13}{6} \pi \right) = 3\sqrt{3}k$$

$$R_y = k \left(\sin \frac{\pi}{2} + 2 \sin \frac{5}{6} \pi + 3 \sin \frac{7}{6} \pi + 4 \sin \frac{9}{6} \pi + 5 \sin \frac{11}{6} \pi + 6 \sin \frac{13}{6} \pi \right) = -3k$$

Therefore, the magnitude of the resultant

$$= \sqrt{(3\sqrt{3}k)^2 + (-3k)^2} = 6k$$

and it makes an angle $\tan^{-1} \left(\frac{-3k}{3\sqrt{3}k} \right) = \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) = \frac{11}{6} \pi$ with OMX .



It is therefore, parallel to the forces $5k$ acting along EF . Further, since all the sides of the hexagon are at a distance OM from O , the algebraic sum of the moments of forces about O

$$= (k + 2k + 3k + 4k + 5k + 6k)OM = 21k.OM$$

Hence the distance of the resultant from the centre

$$= 21k.OM / 6k = \frac{7}{2}OM .$$



Example: Three forces P, Q, R act along the sides of the triangle formed by the lines: $x + y = 1, y - x = 1, y = 2$, taken in order. Find the equation of the line of action of their resultant.

Solution: Let the equation of line of action of the resultant force is

$$xR_y - yR_x = G$$

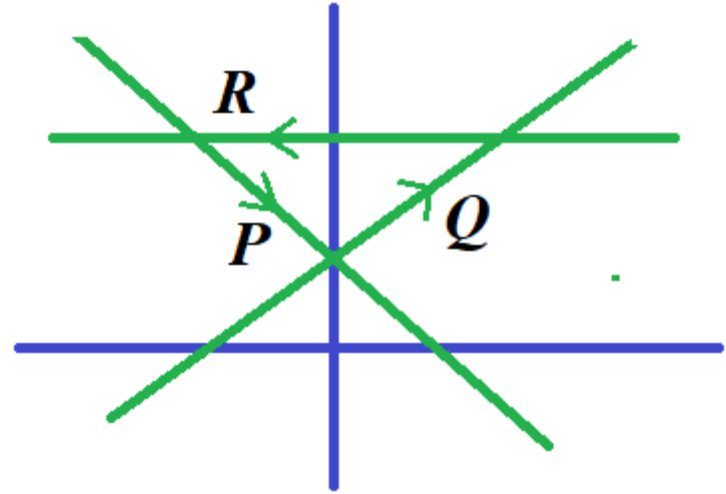
$$\text{We have } R_x = P \cos(-45^\circ) + Q \cos 45^\circ - R \cos 0^\circ = (P + Q - \sqrt{2}R) \frac{1}{\sqrt{2}}$$

$$R_y = P \sin(-45^\circ) + Q \sin 45^\circ - R \sin 0^\circ = (-P + Q) \frac{1}{\sqrt{2}}$$



$$G = -\frac{P}{\sqrt{2}} - \frac{Q}{\sqrt{2}} + 2R$$

$$= (-P - Q + 2\sqrt{2}R) \frac{1}{\sqrt{2}}$$



Therefore the required equation of the resultant is

$$x(-P + Q)/\sqrt{2} - y(P + Q - \sqrt{2}R)/\sqrt{2} = (-P - Q + 2\sqrt{2}R)/\sqrt{2}$$

$$\Rightarrow x(-P + Q) - y(P + Q - \sqrt{2}R) = (-P - Q + 2\sqrt{2}R)$$

$$\Rightarrow (x + y - 1)P + (y - x - 1)Q - (y - 2)\sqrt{2}R = 0$$



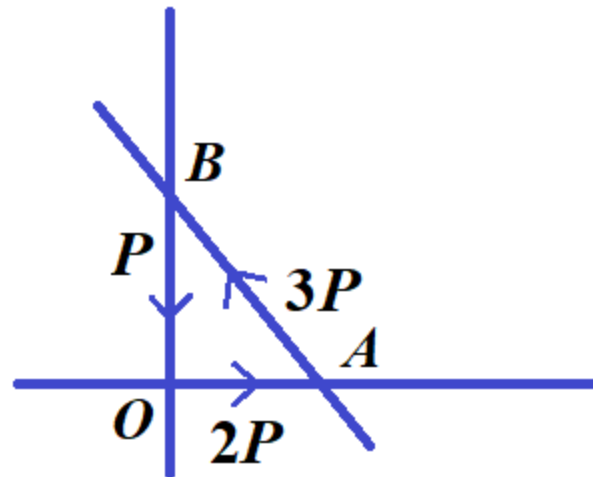
Example: Forces P , $2P$, $3P$ act along the sides of a triangle formed by the lines $x = 0$, $y = 0$ and $3x + 4y = 5$.

Solution:

$$R_x = 2P - 3P\left(\frac{4}{5}\right) = -\frac{2}{5}P$$

$$R_y = -P + 3P\frac{3}{5} = \frac{4}{5}P$$

$$G = 3P \cdot 1 = 3P$$



$$R = \frac{2}{5}\sqrt{5}P, \quad xR_y - yR_x = G \Rightarrow 4x + 2y = 15$$

Example: Forces proportional to 1, 2, 3 and 4 act along the sides AB , BC , AD and DC respectively of a square $ABCD$ the length of whose sides is 2 ft. Find the magnitude and line of action of their resultant.

Example: Forces of relative magnitude 5, 1, 1, -3 act along the side AB , BC , CD , DA respectively of a square $ABCD$. Show that the system is equivalent to a single force and illustrate in a diagram its magnitude, direction and line of action.



Example: $ABCDEF$ is a regular hexagon and O is its centre. Forces of magnitude 1, 2, 3, 4, 5, 6 acts in the lines AB , CB , CD , ED , EF , and AF in the sense indicated by the order of the letters. Reduce the system to a force at O and a couple, and find the point in AB through which the single resultant passes.



General conditions of equilibrium:

- (1) A system of forces in a plane is in equilibrium if the algebraic sums of the resolved parts in any two perpendicular directions vanish and if the algebraic sum of the moments about any point also vanishes.

$$\text{i.e., } R_x = 0, R_y = 0, G = 0.$$

- (2) A system of forces in a plane is in equilibrium if the algebraic sums of the moments of all the forces with respect to each of three non-collinear points is zero.



Without loss of generality, let the three points be $(0,0)$, (ξ,η) , (ξ',η') .

$$G = 0$$

$$G' = G - \xi R_y + \eta R_x = 0$$

$$G'' = G - \xi' R_y + \eta' R_x = 0$$

Since (ξ,η) and (ξ',η') are non-collinear, $R_x = R_y = 0$.



Example: Two systems of forces P, Q, R and P', Q', R' act along the sides BC, CA, AB of a triangle ABC , prove that their resultants will be parallel if $(QR' - Q'R)\sin A + (RP' - R'P)\sin B + (PQ' - P'Q)\sin C = 0$.

Solution: For the first system the sum resolved parts of the forces along the axes over

$$X = P - Q\cos C - R\cos B \text{ and } Y = Q\sin C - R\sin B,$$

where x -axis is taken along BC and y -axis is taken perpendicular to BC with B as origin.



Similarly, for the end system

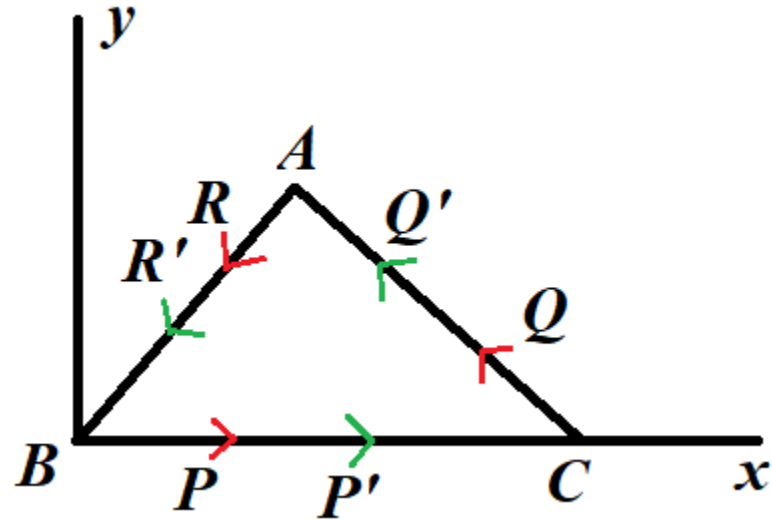
$$X' = P' - Q' \cos C - R' \cos B \text{ and}$$

$$Y' = Q' \sin C - R' \sin B$$

The resultants are to be parallel if

$$\frac{Y}{X} = \frac{Y'}{X'} \Rightarrow YX' = XY'$$

$$\begin{aligned} \text{i.e., } (Q \sin C - R \sin B)(P' - Q' \cos C - R' \cos B) \\ = (P - Q \cos C - R \cos B)(Q' \sin C - R' \sin B) \end{aligned}$$



$$\begin{aligned}
&\Rightarrow P'Q \sin C - QQ' \cos C \sin C - QR' \cos B \sin C - P'R \sin B \\
&\quad - Q'R \cos C \sin B + RR' \cos B \sin B \\
&= PQ' \sin C - PR' \sin B - QQ' \cos C \sin C + QR' \cos C \sin B \\
&\quad - Q'R \cos B \sin C + RR' \cos B \sin B \\
&\Rightarrow (PQ' - P'Q) \sin C + (RP' - R'P) \sin B + QR' (\sin B \cos C + \cos B \sin C) \\
&\quad - Q'R (\sin B \cos C + \cos B \sin C) = 0 \\
&\Rightarrow (PQ' - P'Q) \sin C + (RP' - R'P) \sin B + (QR' - Q'R) \sin (B + C) = 0 \\
&\Rightarrow (PQ' - P'Q) \sin C + (RP' - R'P) \sin B + (QR' - Q'R) \sin A = 0
\end{aligned}$$

Hence the results.



Example: The moments of a system of coplanar forces about three collinear points A, B, C in the plane are G_1, G_2, G_3 . Prove that $G_1BC + G_2CA + G_3AB = 0$.

Solution: Without loss of generality, we can consider the point A be the origin and the line ABC be the x -axis. Also let the system of forces be reduced to a force with components R_x and R_y along the axes and a couple of moment G . Let the coordinates of B and C be $(b,0)$ and $(c,0)$.



Hence, we have

$$G_1 = G \quad \text{for point } A$$

$$G_2 = G - bR_y \quad \text{for point } B$$

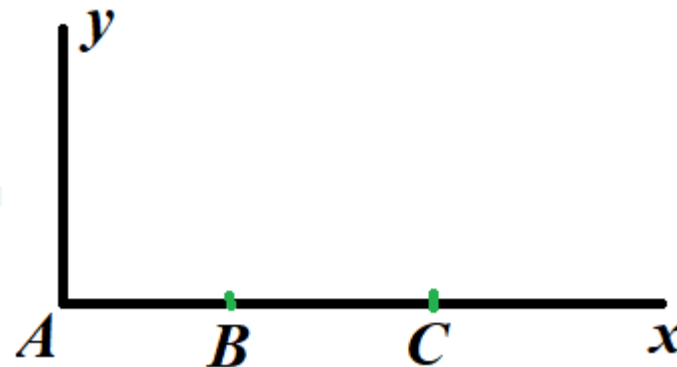
$$G_3 = G - cR_y \quad \text{for point } C$$

Now

$$G_1 BC + G_2 CA + G_3 AB$$

$$= G(c - b) + (G - bR_y)(-c) + (G - cR_y)b \quad [\text{as } BC = c - b, CA = -c, AB = b]$$

$$= Gc - Gb - Gc + bcR_y + Gb - bcR_y = 0. \quad \text{Hence the result is proved.}$$



Example: A system of forces in the plane of a triangle ABC is equivalent to a single force at A , acting long the internal bisector of the angle BAC , and a couple of moment G_1 . If the moments of forces about B and C are G_2 and G_3 respectively, then show that $(b+c)G_1 = bG_2 + cG_3$, where $b=AC$ and $c=AB$

Solution: Without the loss of generality, we can take A as the origin, AB as the x axis and y axis is perpendicular to AB as shown in the figure. It is given that the single resultant, say, R is along the internal bisector of angle A .



$$R_x = R \cos \frac{A}{2} \quad \text{and} \quad R_y = R \sin \frac{A}{2}$$

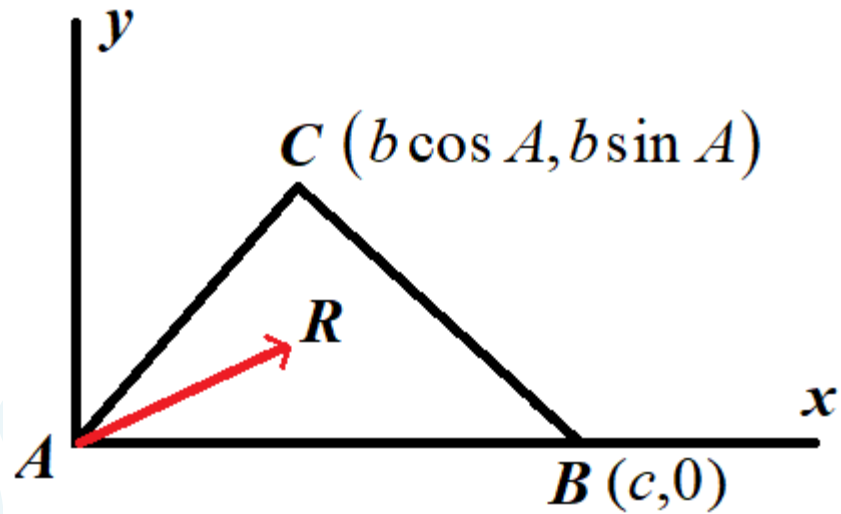
Also, we have $AB=c$, $CA=b$.

The coordinate of B is $(c,0)$ and that of C is $(b \cos A, b \sin A)$

Now according to the given condition we have

$$G_2 = G_1 - cR \sin \frac{A}{2}$$

$$G_3 = G_1 - (b \cos A) R \sin \frac{A}{2} + (b \sin A) R \cos \frac{A}{2}$$



$$\text{Now } bG_2 + cG_3 = (b+c)G_1 - bcR \sin \frac{A}{2} + Rbc \left[\sin A \cos \frac{A}{2} - \cos A \sin \frac{A}{2} \right]$$

$$\text{i.e., } bG_2 + cG_3 = (b+c)G_1 - bcR \sin \frac{A}{2} + bcR \sin \left(A - \frac{A}{2} \right)$$

$$\Rightarrow bG_2 + cG_3 = (b+c)G_1 \quad (\text{Proved})$$



Part – II

Virtual Work



In statics we always have a body or a system of bodies in equilibrium under the action of given forces. The general method of solving problems is write down the equation of conditions of equilibrium, i.e., equations of resolution and equation of moment. But, a great disadvantage is that such equations contain unknown forces, like reaction, thrust, tension etc which should be eliminated.

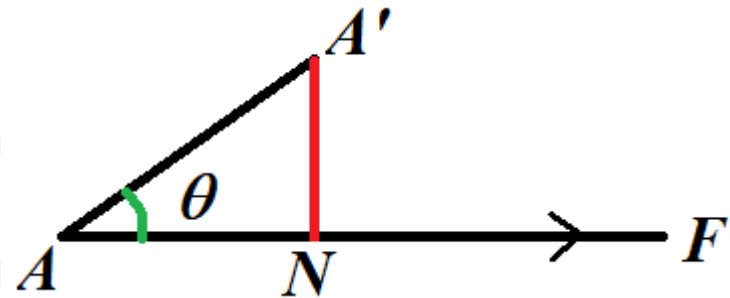
The principle of virtual work affords us a method of writing down equations involving no such unknown forces or, involving only such unknown forces which we want to find out. It is a powerful method and may be used with great advantage.



Work done by a force:

If a force $\vec{F}$ acts at a point of a body and the point of application A' undergoes a displacement AA' then the product $F \cdot AA' \cos \theta$ is called the work

done by the force $\vec{F}$ in the displacement AA' , where θ is the angle between the direction of force and displacement AA' and $|\vec{F}| = F$.



Let $A'N$ be drawn perpendicular to the line of action of $\vec{F}$. Then AN is the orthogonal projection of AA' on the line of action of $\vec{F}$ and is equal



to $AA' \cos \theta$ and the work done is $F.AN$. So, the work done by a force in any displacement is equal to the product of the force and orthogonal projection of the displacement on the line of action of the force. AN is called the displacement in the direction of force.

Again $F AA' \cos \theta = AA'.F \cos \theta$. So, the work done in any displacement is also equal to the product of the displacement and the resolve part of the force in the direction of the displacement. No work is done when the displacement is perpendicular to the force.

It may be noted that the work done by the force $\vec{F}$ in a displacement vector $\overrightarrow{AA'}$ is the scalar quantity $\vec{F} \cdot \overrightarrow{AA'}$.



Virtual Work:

When the point of application A of a force $\vec{F}$ does not actually move to A' , but it is imagined to move A' then the displacement AA' is called a virtual displacement of the point A and the corresponding work done by the force $\vec{F}$ on the assumption that the point of application has undergone the displacement AA' is called the virtual work done by the force $\vec{F}$ in the virtual displacement AA' .

It is of course measured by $F AA' \cos \theta$ or $F \cdot AN$ or $AA' \times$ resolved part of $\vec{F}$ in the direction of AA' .



Static bodies are at rest, so under the action of forces those do not actually move. All displacements of the points of application of the forces are therefore hypothetical, i.e., imagined and the works calculated are the works which would be done if the displacements were actually made, that is why the displacements and the works done are called Virtual.

Theorem: When any number of forces act on a particle and the particle under goes a small displacement then the total work done by all the forces is equal to the work done by their resultant.



Principle of Virtual work for a system of coplanar forces acting on a particle:

The necessary and sufficient condition that a particle acted upon a number of coplanar forces, be in equilibrium is that the algebraic sum of the virtual works done by the forces in a small displacement, consistent with the geometrical conditions of the system, is zero.



Principle of virtual work for a system of a coplanar forces acting at different points of rigid body:

The necessary and sufficient condition that a rigid body acted upon by a number of coplanar forces, be in equilibrium, is that the algebraic sum of the virtual works done by the forces in a small displacement, consistent with the geometrical condition of the system, is zero.

Proof: Let us take Ox , Oy as the axes of the coordinates in the plane of the forces.

The most general virtual displacement of the body, the plane of the forces remaining unaltered is a rotation about O together with a translation.



Let the body be given a rotation about O through any small angle α and any small translation whose components parallel to OX and OY are a, b .

Let $P_1, P_2, P_3, \dots$ be the forces acting at the points $A_1(x_1, y_1), A_2(x_2, y_2), A_3(x_3, y_3), \dots$ of the body and let its components parallel to axes be $X_1, Y_1; X_2, Y_2; X_3, Y_3; \dots$ respectively.

Again, let the polar coordinates of $A_1, A_2, A_3, \dots$ be $(r_1, \theta_1), (r_2, \theta_2), (r_3, \theta_3), \dots$ respectively.

Then $x_1 = r_1 \cos \theta_1, y_1 = r_1 \sin \theta_1$

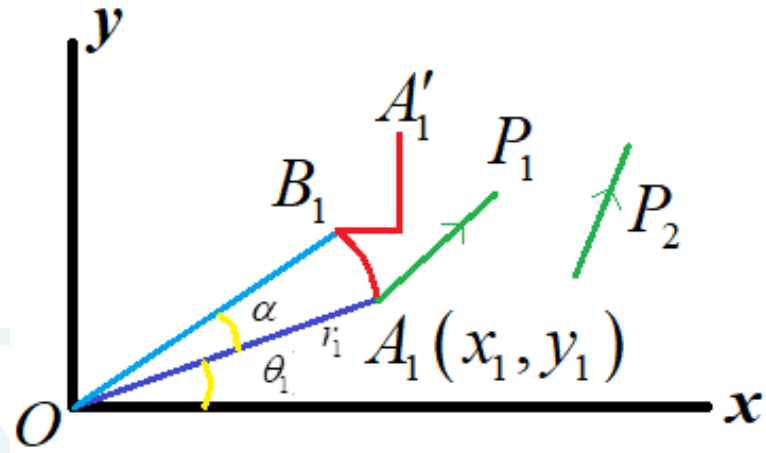


Due to the rotation A_1 and B_1 whose polar coordinates are $(r_1, \theta_1 + \alpha)$. So its Cartesian coordinates are $(r_1 \cos(\theta_1 + \alpha), r_1 \sin(\theta_1 + \alpha))$

The point A'_1 is obtained by a translation of point B'_1 whose components are a, b parallel to OX and OY respectively.

Therefore, the coordinates of A'_1 are

$$(r_1 \cos(\theta_1 + \alpha) + a, r_1 \sin(\theta_1 + \alpha) + b).$$



If δx_1 and δy_1 are the changes in the coordinates of A_1 due to its displacement from A_1 to A'_1 then

$$\begin{aligned}\delta x_1 &= r_1 \cos(\theta_1 + \alpha) + a - x_1 \\ &= r_1 \cos \theta_1 \cos \alpha - r_1 \sin \theta_1 \sin \alpha + a - x_1 \\ &= x_1 \cos \alpha - y_1 \sin \alpha + a - x_1 \\ &= x_1 - y_1 \alpha + a - x_1 \\ &= a - \alpha y_1\end{aligned}$$

[Writing $\alpha = \sin \alpha$ and $1 = \cos \alpha$ neglecting higher powers of α which are small quantities as α is small.]



$$\begin{aligned}
\delta y_1 &= r_1 \sin(\theta_1 + \alpha) + b - y_1 \\
&= r_1 \sin \theta_1 \cos \alpha + r_1 \cos \theta_1 \sin \alpha + b - y_1 \\
&= y_1 \cos \alpha + x_1 \sin \alpha + b - y_1 \\
&= y_1 + x_1 \alpha + b - y_1 \\
&= b + \alpha x_1
\end{aligned}$$

Therefore, the virtual work done by force P_1 is $X_1 \delta x_1 + Y_1 \delta y_1$

$$\begin{aligned}
&= X_1 (a - \alpha y_1) + Y_1 (b + \alpha x_1) \\
&= aX_1 + bY_1 + \alpha (x_1 Y_1 - y_1 X_1)
\end{aligned}$$



Similarly, we can obtain the virtual work of the forces $P_2, P_3, \dots$. Where a, b and α are same for each forces.

Hence the (algebraic sum of the) virtual work done by all the forces is

$$\begin{aligned} & \text{equal to } \sum aX_1 + \sum bY_1 + \sum \alpha(x_1Y_1 - y_1X_1) \\ & = a \sum X_1 + b \sum Y_1 + \alpha \sum (x_1Y_1 - y_1X_1) \end{aligned} \quad (1)$$

Now let the system be in equilibrium then we have

$$\sum X_1 = 0, \sum Y_1 = 0, \sum (x_1Y_1 - y_1X_1) = 0$$

Hence from (1) it follows that the algebraic sum of the virtual works done by the system of forces is zero. (Thus the necessary part is proved)



Conversely, let the algebraic sum of the virtual work done by the system of forces is zero for every arbitrary small displacement in the plane of forces.

Then we have from (1)

$$a \sum X_1 + b \sum Y_1 + \alpha \sum (x_1 Y_1 - y_1 X_1) = 0 \quad (2)$$

for all arbitrary small values of a , b and α

Let us consider a displacement such that the body is moved only through a displacement parallel to x -axis.

Then $a \neq 0, b = 0, \alpha = 0$ and from (2) we have

$$a \sum X_1 = 0 \Rightarrow \sum X_1 = 0 \text{ as } a \neq 0.$$



Similarly, choosing a displacement parallel to the y -axis it can be shown that $\sum Y_1 = 0$

Finally, let the displacement be one of a simple rotation.

Then $a = 0, b = 0, \alpha \neq 0$ and (2) gives

$$\alpha \sum (x_1 Y_1 - y_1 X_1) = 0 \Rightarrow \sum (x_1 Y_1 - y_1 X_1) = 0 \text{ as } \alpha \neq 0$$

Then $\sum X_1 = 0, \sum Y_1 = 0$ and $\sum (x_1 Y_1 - y_1 X_1) = 0$, which implies that the system is in equilibrium. (Hence the sufficient part is proved)



Unwanted Forces:

If we want to get rid of a certain unwanted force, we imagine a displacement (consistent with the geometrical condition of the system) at the right angle to the line of action of that force.

Tension and Thrust:

The tension in a string connecting two particles (or bodies) has a tendency to bring the particles (or bodies) together. Therefore, if a string be considered as divided at any point P of it into two parts AP and PB , the



tension at P in the parts AP and PB are directed respectively towards B and A .

Just the reverse is the case with thrust applied by a rod which keeps the two particles (or bodies) at its ends from coming together. Consequently, the thrust at C point in the two parts AP and PB , say, in which a rod is divided at P , are directed towards A and B , respectively.

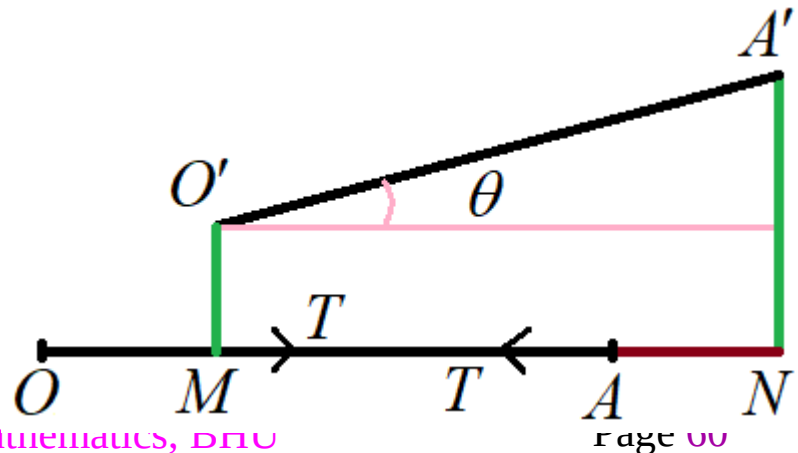


Forces which do not appear in the equation of virtual work:

There are certain forces which have no contribution to the virtual work i.e. the virtual works done by them are zero. Hence these forces may be omitted in forming the equation of virtual work. We state some usually occurring such forces.

(1) The tension of an inextensible string:

Let OA be such a string whose tension is T . In the displaced position, $O'A'$ (let) is the string and we draw perpendicular $O'M$ and $A'N$.



Hence the virtual work of the tension.

$$= T.OM - T.AN = T(OM - AN) = T(OA - MN)$$

$$[OM - AN = (OA - MA) - (MN - MA) = OA - MN]$$

$$= T.OA(1 - \cos \theta)$$

$$= T.OA(1 - 1) = 0 \quad (\text{to the first order of small quantities})$$

(2) The tension thrust in a rod whose length remains unaltered:

(3) Forces of action and reaction between two particles when the distance between them remains in variables:

Similar to inextensible string.



(4) The reaction R at any smooth surface with which the body is in contact:

If the surface is smooth, the reaction R is normal to the surface at the point of contact P , so that if P moves to a neighboring point P' , then PP' is at right angle to the force. Its virtual work is thus zero.

If the surface is rough then the work done by friction F viz, $F(-PP')$ comes in the equation since it is not, in general, zero.



(5) The reaction at any point P with a fixed surface on which the body rolls without sliding:

For the point of contact P of the body is for the moment at rest and so its displacement is zero.

The normal reaction at P and the friction at P have no displacement and hence zero virtual work.

It is evident that whether a force will appear in the equation of virtual work or not depends not only on the force but also on the displacement given.



Forces which appear in the equation of virtual work

Every force which does not work due to the displacement imagined will appear in the equation of virtual work.

(1) The tensions at the ends of a string or rod when the displacement alters its length:

If T is the equal tensions and l is the length of the string or rod, then the work done is $-T\delta l$

Proof: Let OA be such a string whose tension is T . In the displaced position let $O'A'$ be the string and we draw perpendicular $O'M$ and $A'N$ on OA .



Let us suppose that the length of string or rod l be not constant.

Let in the displaced position $O'A' = OA + \delta(OA) = l + \delta l$

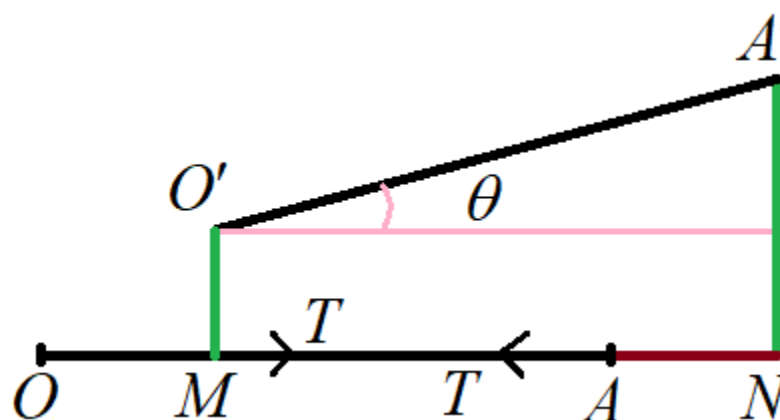
Then the total work done by the tension T at O and T at A is equal to

$$T \cdot OM + (-T) \cdot AN$$

$$= T(OA - MA) - T(MN - MA)$$

$$= T(OA - MN)$$

$$= T(OA - O'A' \cos \theta) = T(OA - O'A') \quad [\cos \theta \cong 1 \text{ as } \theta \text{ is small}]$$



$$= T(l - l - \delta l) = -T\delta l$$

= – Tension in the string or rod $\times$ increment in the length of the string or rod.

(2) The thrust at the ends of a rod when displacement alters its length:

If T be the equal thrust and l be the length of the rod, work done is $T\delta l$

Proof: Same as above

(3) Weight of a body or of a system of bodies:

If W be the weight and y be the depth of the centre of gravity of the body or of the system of bodies below some fixed plane, then work done by the weight due to a small displacement δy be $W\delta y$, but if y is the height of



the centre of gravity above some fixed plane then the work done is $-W\delta y$. For the distance of the centre of gravity where the weight is measured positively in a direction opposite to the direction in which the weight acts.



Example: Four equal heavy uniform rods are freely joined so as to form a rhombus which is freely suspended by one angular point and the middle points of the two upper rods are connected by a light rod, so that the rhombus cannot collapse. Prove that the thrust of this rod is $4W \tan \alpha$, where W is the weight of each rod and 2α is the angle of the rhombus at the point of suspension.

Solution:

Let $ABCD$ be the rhombus. Again, let E, F, M, N be the midpoint of AB, BC, CD, DA respectively. Let the length of each rod be a .

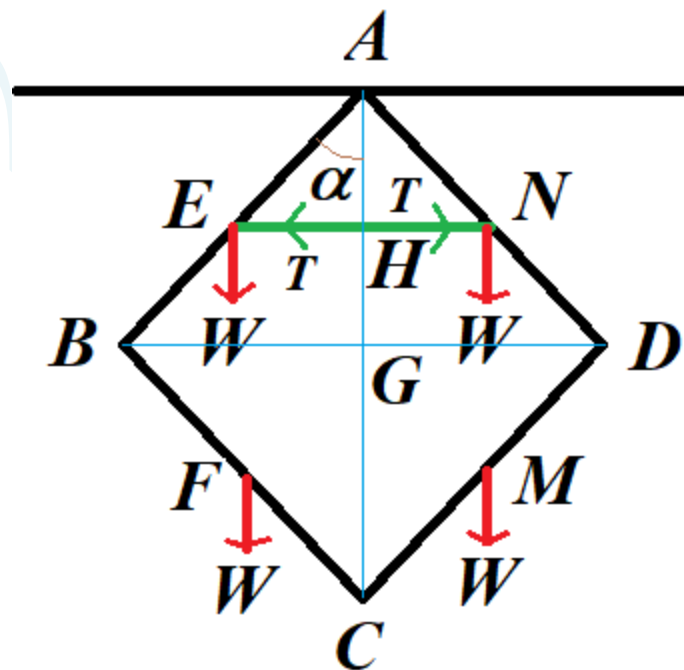


Since $ABCD$ is a rhombus we have G as the midpoint of AC and BD and also G is the centre of gravity of the system.

So the force $4W$ acts at G downwards with W being the weight of each rod.

Let T be the thrust of the rod.

Let the system be given a virtual displacement so that the angle α changes.



Then the equation of virtual work is

$$4w\delta(AG) + T\delta(EN) = 0 \quad (1)$$

Now from right angle triangle $\triangle ABG$ we have

$$\cos \alpha = \frac{AB}{AG} = \frac{a}{AG} \Rightarrow AG = a \cos \alpha.$$

From $\triangle AEH$ we get $EH = \frac{a}{2} \sin \alpha$. So, $EN = 2EH = a \sin \alpha$.

Therefore, From (1) we have

$$4W\delta(a \cos \alpha) + T\delta(a \sin \alpha) = 0 \Rightarrow -4Wa \sin \alpha \delta \alpha + Ta \cos \alpha \delta \alpha = 0$$

$$\Rightarrow a(T \cos \alpha - 4W \sin \alpha) \delta \alpha = 0$$

$$\Rightarrow T \cos \alpha - 4W \sin \alpha = 0 \quad [\text{as } \delta \alpha \neq 0, a \neq 0] \Rightarrow T = 4W \tan \alpha.$$



Example: Four equally uniform rods are jointed to form a rhombus $ABCD$ which is placed in a vertical plane with AC vertical and A resting on horizontal plane. The rhombus is kept in the position in which $\angle BAC$ is θ by a light string joining B and D , show that its tension is $2w \tan \theta$, where w is the weight of the rod.

Example: Four equal uniform rods each of weight W , are jointed to form a rhombus $ABCD$, which is placed in a vertical plane with AC vertical and A resting on a horizontal plane. The rhombus is kept in the position in



which $\angle BAC = \theta$ by a light string joining B and D . Find the tension of the string.

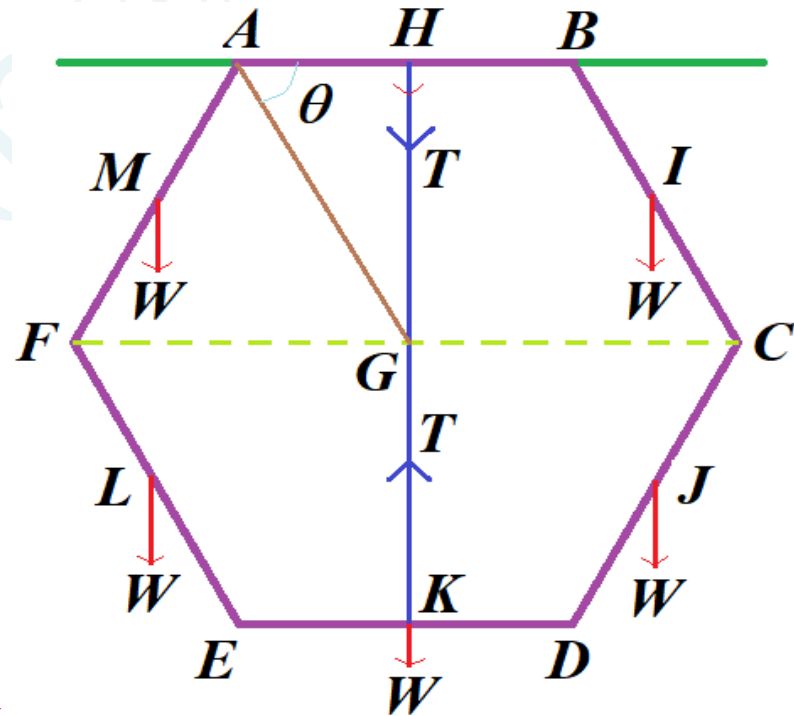
Example: Four uniform rods are freely jointed at their extremities and form a parallelogram $ABCD$ which is suspended by the joint A and kept in shape by a string AC . Prove that the tension of the string is equal to half the whole weight.



Example: Six equal rods AB, BC, CD, DE, EF and FA are each of weight W and are freely jointed at their extremities so as to form a hexagon; the rod AB is fixed in a horizontal position and the middle points of AB and DE are joined by a string. Prove that the tension of the string is $3W$.

Solution: Let H, I, J, K, L, M be the midpoints of AB, BC, CD, DE, EF and FA respectively.

Let T be the tension of the string.



Let at the position of equilibrium $\angle FAB = 2\theta$ and let G be the centre of gravity of system. Now the force $6W$ acts at G downwards and the depth of G from the fixed rod HG . Also the length of string is $HK=2HG$.

Now we give a small virtual displacement so that the angle θ changes.

Then the equation of virtual work is

$$6W\delta(HG) - T\delta(2HG) = 0$$

$$\Rightarrow (6W - 2T)\delta(HG) = 0$$

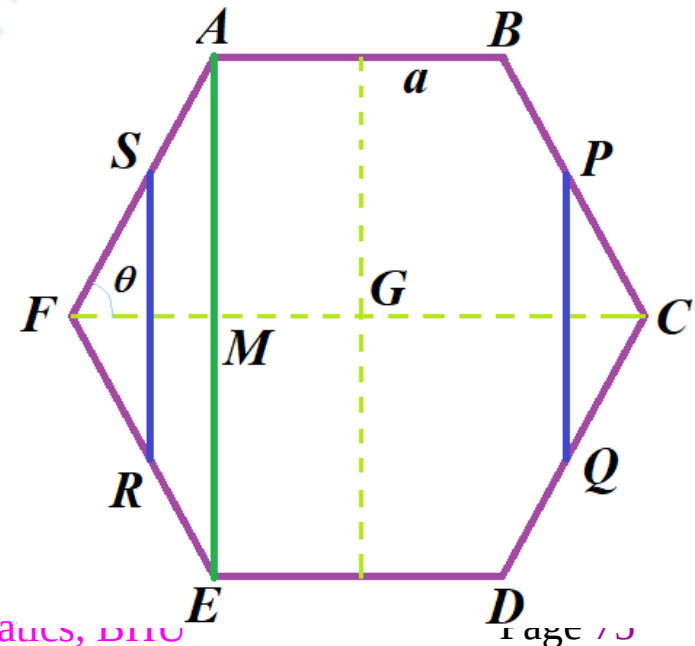
$$\Rightarrow 2T = 6W \text{ as } \delta(HG) \neq 0$$

$$\Rightarrow T = 3W \text{ (Proved)}$$



Example: Six equal bars are freely jointed at their extremities forming a regular hexagon $ABCDEF$ which is kept in shape by vertical strings jointed in the middle points of BC , CD and AF , FE respectively and AB is fixed horizontally. Show that the tension of each string is three times the weight of a bar.

Solution: Let W be the weight and a be the length of each rod. Let P , Q , R , S be the mid points of BC , CD , EF and FA respectively.



Let T be the tension of each of the string PQ and RS .

Let at equilibrium position $\angle AFE = 2\theta$

So $\angle AFG = \angle EFG = \theta$, G being the centre of gravity of the system.

Now, from $\triangle AFM$ we have

$$AM = AF \sin \theta = a \sin \theta$$

And from $\triangle AEF$ we have

$$SR = \frac{1}{2} AE = AM = a \sin \theta$$

Now, the depth of a G from the horizontal bar AB is $AM = a \sin \theta$



Now, the force $6W$ acts at G downwards. We give a small virtual displacement so that the angle θ alters.

Then the equation of virtual work is

$$6W\delta(a\sin\theta) - 2T(a\sin\theta) = 0$$
$$\Rightarrow 6W - 2T = 0 \text{ [as } a\sin\theta \neq 0] \Rightarrow T = 3W.$$

Hence, the tension of each string is three times the weight of a bar.

Example: Six equal heavy beams are free by jointed at their ends to form a hexagon and are placed in a vertical plane with one beam resting on a horizontal plane; the middle points of the two upper slant beams, which



are inclined at an angle θ to the horizon, are connected by a light rod. Find its tension in terms of W and θ , where W is the weight of each beam.

Example: A regular hexagon $ABCDEF$ is composed of six equal heavy rods freely jointed together and two opposite angles C and F are connected by a string, which is horizontal, AB being contact with a horizontal plane. A weight W is placed at the middle point DE . If w be the weight of each rod. Show that the tension of the string is $\frac{3w+W}{\sqrt{3}}$.



Example: A uniform rod of length $2a$ rests in equilibrium against a smooth vertical wall and upon a smooth peg distant b from the wall. Show by the principle of virtual work that, in the position of equilibrium, the rod is inclined to the wall at an angle $\sin^{-1}(b/a)^{\frac{1}{3}}$.

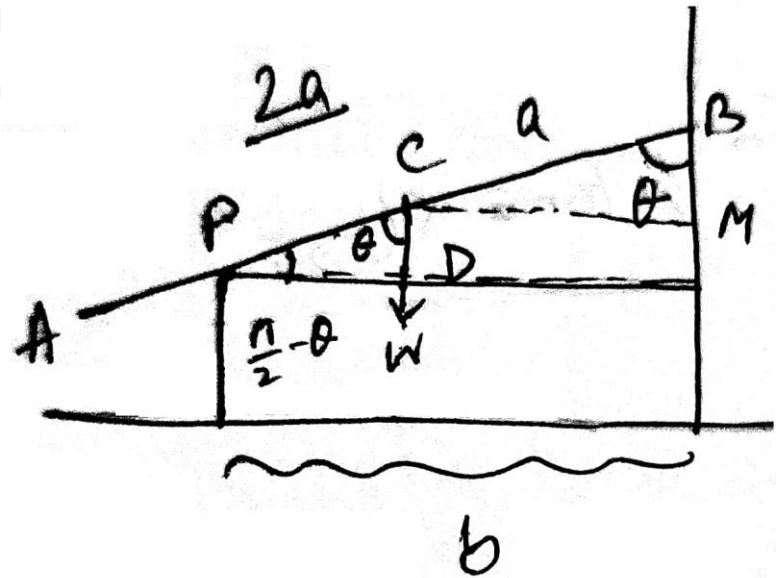
Solution:

The equation of virtual work is

$$-W\delta(CD) = 0$$

We have, $CM = a \sin \theta$

Therefore, $PD = b - a \sin \theta$



Now $CD = PD \cot \theta = (b - a \sin \theta) \cot \theta$

So we have $W \delta [(b - a \sin \theta) \cot \theta] = 0$

$$\Rightarrow W \delta \theta \{ -a \cos \theta \cot \theta - (b - a \sin \theta) \operatorname{cosec}^2 \theta \} = 0$$

$$\Rightarrow -a \cos \theta \cot \theta - (b - a \sin \theta) \operatorname{cosec}^2 \theta = 0$$

$$\Rightarrow a \frac{\cos^2 \theta}{\sin \theta} + (b - a \sin \theta) \frac{1}{\sin^2 \theta} = 0$$

$$\Rightarrow a(1 - \sin^2 \theta) \sin \theta + (b - a \sin \theta) = 0$$

$$\Rightarrow a \sin \theta - a \sin^3 \theta + b - a \sin \theta = 0$$

$$\Rightarrow \sin^3 \theta = b/a \quad \Rightarrow \theta = \sin^{-1} (b/a)^{\frac{1}{3}}.$$



Example: A heavy uniform rod of length $2a$, rests with its ends in contact with two smooth inclined plane, of inclination α and β to the horizon. If θ be the inclination of the rod to the horizon prove, by the principle of virtual work, that

$$\tan \theta = \frac{1}{2}(\cot \alpha - \cot \beta).$$

Example: $ABCD$ is a rhombus formed with four rods each of length l and negligible weight joined by smooth hinges. A weight W is attached to the lowest hinge C and the frame rests on two smooth pages in horizontal line



in contact with the rods AB and AD . B and D are horizontal line and are joined by a string. If the distance of the pegs is $2c$ and the angle at A is 2α , show that the tension in the string is $W \tan \alpha \left(\frac{c}{2l} \operatorname{cosec}^3 \alpha - 1 \right)$.

Example: A rhombus is formed of four rods each of weight W and length l with smooth joints. It rests symmetrically with its upper sides in contact with two smooth pegs at the same level and at a distance $2a$ apart. A weight W' is hung to the lowest point. If the sides of the rhombus make an angle θ with the vertical, prove that $\sin^3 \theta = \frac{a(4W + W')}{l(4W + 2W')}$.



Solution: We have equation of virtual work is

$$4W\delta(LG) + W'\delta(LC) = 0 \quad (1)$$

$$LG = l \cos \theta - a \cot \theta \text{ and } LC = 2l \cos \theta - a \cot \theta.$$

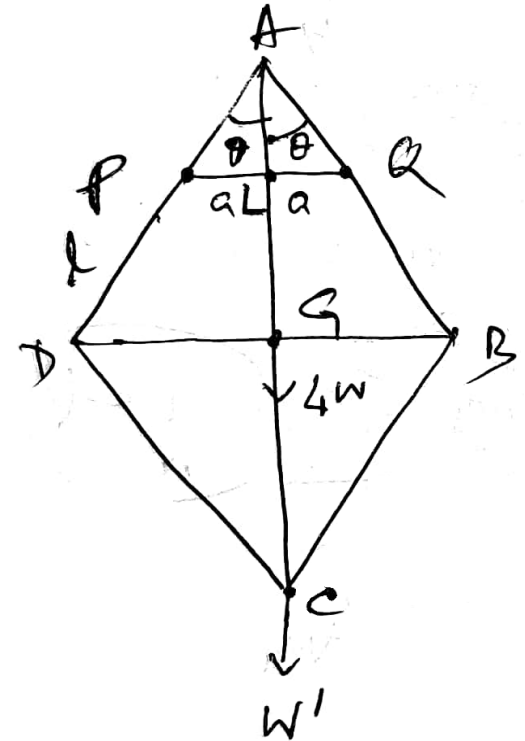
From (1) we have

$$4W\delta(l \cos \theta - a \cot \theta) + W'\delta(2l \cos \theta - a \cot \theta) = 0$$

$$\Rightarrow 4W(-l \sin \theta + a \operatorname{cosec}^2 \theta) + W'(-2l \sin \theta + a \operatorname{cosec}^2 \theta) = 0$$

$$\Rightarrow -l(2W' + 4W) \sin \theta + a(4W + W') \operatorname{cosec}^2 \theta = 0$$

$$\Rightarrow \sin \theta (4W + 2W') l = \operatorname{cosec}^2 \theta (4W + W') a$$



$$\Rightarrow \sin^3 \theta = \frac{a(4W + W')}{l(4W + 2W')}.$$

Example: A square of side $2a$ is placed with its plane vertical, between two smooth pegs which one in a horizontal line and at a distance c apart. Show that it will be in equilibrium when the inclination of one of its edges

to the horizon is either $\frac{\pi}{4}$ or $\frac{1}{2} \sin^{-1} \frac{a^2 - c^2}{c^2}$.



Hooke's Law

If the natural and expended length of an elastic string are a and l and the modulus of elasticity is λ then by Hooke's law the tension of the string, T

is given by $T = \lambda \frac{l - a}{a}$

Example: A heavy elastic string, whose natural length is $2\pi a$, is placed round a smooth cone whose axis is vertical and whose semi-vertical angle is α . If W be the weight and λ the modulus of elasticity of the string, prove that it will be in equilibrium when in the form of a circle of radius



$$a \left(1 + \frac{W}{2\pi\lambda} \cot \alpha \right).$$

Solution: Let ABO be the smooth cone.

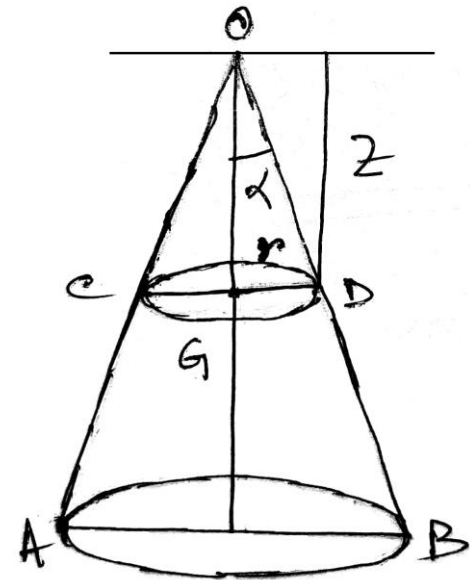
Let at the position of equilibrium. Let CD be the elastic string with centre at G and T be the tension on the string.

Let r be the radius of the string and z be the depth of the centre below O , the vertex of the cone.

Then $r = z \tan \alpha$.

Therefore, the length of the string

$$l = 2\pi r = 2\pi z \tan \alpha$$



Here the force W (the weight of the string) acts at G downwards and the tension of the string work.

We now give a small virtual displacement in which the string moves vertically downwards parallel to itself remaining with contact of the cone.

The equation of virtual work is

$$W\delta z - T\delta l = 0 \Rightarrow W\delta z - T\delta(2\pi z \tan \alpha) = 0$$

$$\Rightarrow W\delta z - T2\pi \tan \alpha \delta z = 0 \Rightarrow W - T2\pi \tan \alpha = 0$$

$$\Rightarrow T = \frac{W}{2\pi} \cot \alpha \quad (1)$$

Also by Hooke's law we have



$$T = \lambda \frac{2\pi r - 2\pi a}{2\pi a} = \lambda \frac{r - a}{a} \quad (2)$$

From (1) and (2) we get

$$\begin{aligned} \lambda \frac{r - a}{a} &= \frac{W}{2\pi} \cot \alpha \Rightarrow r - a = \frac{aW}{2\pi\lambda} \cot \alpha \\ \Rightarrow r &= a \left(1 + \frac{W}{2\pi\lambda} \cot \alpha \right). \end{aligned}$$

Example: One end of a uniform rod AB of length a and weight W , is attached by a frictionless joint to a smooth vertical wall and the other end B is smoothly jointed to an equal rod BC . The middle points of the rod,



are jointed by an elastic string, of natural length a and modulus of elasticity $4W$. Prove that the system can rest in equilibrium in a vertical plane with C in contact with the wall below A and the angle between the rods is $2 \sin^{-1} \left(\frac{3}{4} \right)$.



Part – III

General Motion in Two Dimensions



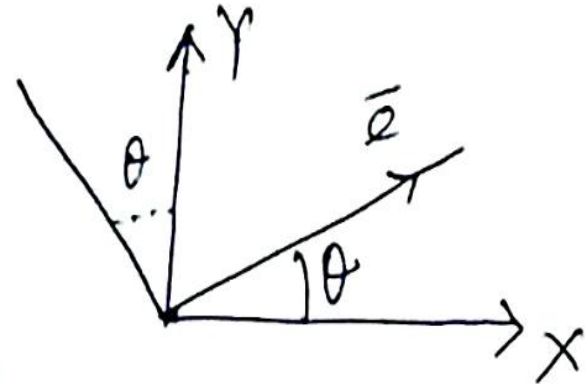
Rotation of a vector

Let a vector $\vec{A}$ be defined in a plane OXY . Let it be rotating about its tip at the origin. Let at any moment (time) its direction makes θ angle with the x -axis OX . Let $\vec{e}$ be unit vector along the direction of $\vec{A}$. The vector changes due to change in its magnitude as well as its direction.

The vector is defined as $\vec{A} = A\vec{e}$.

Now its rate of change with respect to θ may be expressed as

$$\frac{d\vec{A}}{d\theta} = A \frac{d\vec{e}}{d\theta} + \vec{e} \frac{dA}{d\theta}, \quad (1)$$



where on right hand side first term expresses change in the direction, while second term expresses change in its magnitude.

Taking dot product of (1) with the unit vector $\vec{e}$ we have

$$\vec{e} \cdot \frac{d\vec{A}}{d\theta} = A \left(\vec{e} \frac{d\vec{e}}{d\theta} \right) + \frac{dA}{d\theta} \quad (2)$$

$$\text{Now, } \frac{dA}{d\theta} = \frac{1}{2A} \frac{dA^2}{d\theta} = \frac{1}{2A} \frac{d}{d\theta} (\vec{A} \cdot \vec{A}) = \frac{1}{2A} \left[\vec{A} \cdot \frac{d\vec{A}}{d\theta} + \frac{d\vec{A}}{d\theta} \cdot \vec{A} \right]$$

$$\text{i.e., } \frac{dA}{d\theta} = \frac{\vec{A}}{A} \cdot \frac{d\vec{A}}{d\theta} = \vec{e} \cdot \frac{d\vec{A}}{d\theta} \quad (3)$$



Then from (2) and (3) we have

$$\vec{e} \cdot \frac{d\vec{e}}{d\theta} = 0 \text{ as } A \neq 0 \quad (4)$$

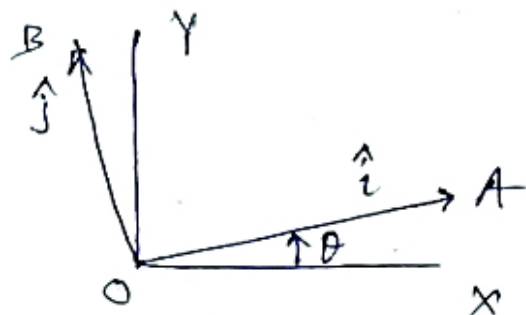
Now we can conclude following two results:

- (1) The magnitude of an unit vector rotating in two dimension about its initial point does not change with respect to θ .
- (2) If $\vec{e}$ be a unit vector, then $\frac{d\vec{e}}{d\theta}$ is another unit vector, but perpendicular to $\vec{e}$.



Now let us apply above results on two mutually perpendicular unit vectors $\hat{i}$ and $\hat{j}$.

Let $\hat{i}$ be along OA and $\hat{j}$ be along OB . Plane OAB be rotating with respect to a fixed plane OXY .



Let at any moment (time), OA makes angle θ with OX , then $\frac{d\hat{i}}{d\theta}$ will represent a unit vector

perpendicular to OA , i.e. along OB . Further $\frac{d\hat{j}}{d\theta}$

will represent a unit vector along OA .



Thus we have $\frac{d\hat{i}}{d\theta} = \hat{j}$ and $\frac{d\hat{j}}{d\theta} = -\hat{i}$,

where negative sign occurs because the sense of change in $\hat{j}$ will be opposite to that in $\hat{i}$.

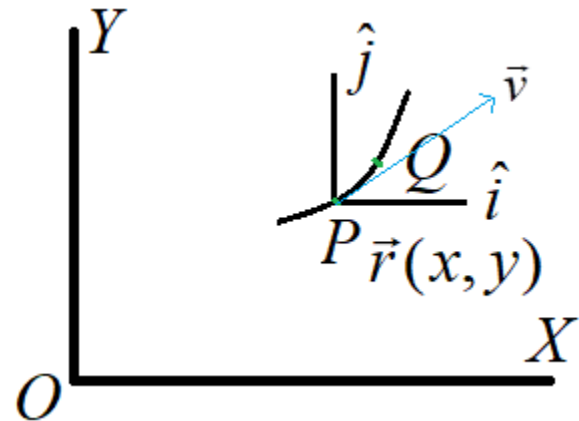


Cartesian Components of Motion

Let OXY be a Cartesian plane a fixed frame of reference. Let a particle describe a path in the plane. Let us consider two positions on the path P and Q very close to each other. Two perpendicular directions at both positions are parallel to x -axis and y -axis respectively.

Thus as the position changes on the path, the two perpendicular directions do not change.

Let position vector of P be $\vec{r}(x, y)$. If $\hat{i}$ and $\hat{j}$ are unit vectors along x -axis and y -axis,



respectively, then we may write $\vec{r} = x\hat{i} + y\hat{j}$

Now, the rate of change with respect to time t with reference to fixed frame will be the linear velocity $\vec{v}$ of the particle and it is given by

$$\dot{\vec{r}} = \dot{x}\hat{i} + \dot{y}\hat{j} + x\dot{\hat{i}} + y\dot{\hat{j}} .$$

But, in Cartesian system unit vector does not change. So we have

$\dot{\hat{i}} = \dot{\hat{j}} = 0$. Thus we have $\dot{\vec{r}} = \dot{x}\hat{i} + \dot{y}\hat{j} = \vec{v}$

Now, if α be angle made by the velocity vector with the positive x -axis, v is the magnitude of the velocity then

$$v^2 = \dot{x}^2 + \dot{y}^2 \text{ and } \dot{x} = v \cos \alpha, \dot{y} = v \sin \alpha, \text{ i.e., } \tan \alpha = \frac{\dot{y}}{\dot{x}} = \frac{dy}{dx} .$$



Similarly, if $\vec{a}$ be acceleration with $|\vec{a}| = a$ and β be the angle between the acceleration vector and positive x-axis, then

$$\vec{a} = \ddot{x}\hat{i} + \ddot{y}\hat{j}, a^2 = \ddot{x}^2 + \ddot{y}^2 \text{ and } \tan \beta = \frac{\ddot{y}}{\ddot{x}}.$$

So, it is clear from above that the direction of the velocity and acceleration at any point on the path are different.



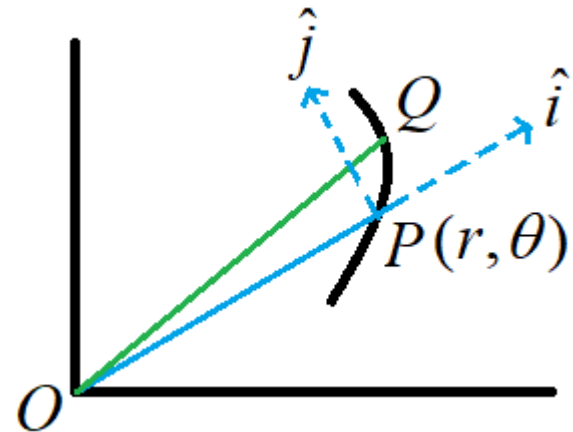
Radial and Transverse Components (Polar Components)

Two perpendicular directions in polar system of a two dimensional frame of reference are radial and transverse directions (cross radial direction).

Let the position of a moving particle at time on its path be $P(r, \theta)$.

Further, let unit vector along the radial direction OP be $\hat{i}$ and that along transverse direction at P along θ increasing direction be $\hat{j}$.

It may be noted that the radial directions at



two very close positions P and Q are not parallel. It means that the radial direction $\hat{i}$ and the transverse direction $\hat{j}$ change with θ .

$$\text{So we have } \frac{d\hat{i}}{d\theta} = \hat{j} \text{ and } \frac{d\hat{j}}{d\theta} = -\hat{i}.$$

Now, the position vector at P be $\vec{r} = r\hat{i}$, where $|\vec{r}| = r$.

Now, its rate of change with time with reference to fixed frame will give the linear velocity of the particle at P along tangent, i.e.,

$$\begin{aligned} \vec{v} &= \frac{d\vec{r}}{dt} = \dot{\vec{r}} = \dot{r}\hat{i} + r\frac{d\hat{i}}{dt} = \dot{r}\hat{i} + r\frac{d\hat{i}}{d\theta}\frac{d\theta}{dt} \\ \vec{v} &= \dot{r}\hat{i} + r\dot{\theta}\hat{j} \end{aligned} \quad (1)$$



So, the two velocity components at $P(r, \theta)$ are

- (i) Radial velocity $\dot{r}$ along the radial direction ($\hat{i}$).
- (ii) Transverse velocity $r\dot{\theta}$ along the transverse direction ($\hat{j}$).

If the velocity vector makes an angle α with the radial direction then

$$\tan \alpha = \frac{r\dot{\theta}}{\dot{r}} = r \frac{d\theta}{dr} \text{ and } v^2 = \dot{r}^2 + (r\dot{\theta})^2, \text{ where } |\vec{v}| = v.$$

Differentiating (1) with respect to time with reference to fixed frame we get

$$\vec{a} = \frac{d\vec{v}}{dt} = \ddot{r}\hat{i} + (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{j} + \dot{r}\frac{d\hat{i}}{dt} + r\dot{\theta}\frac{d\hat{j}}{dt}$$



$$\text{i.e., } \vec{a} = \ddot{r}\hat{i} + (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{j} + r\frac{d\hat{i}}{d\theta}\frac{d\theta}{dt} + r\dot{\theta}\frac{d\hat{j}}{d\theta}\frac{d\theta}{dt}$$

$$\text{i.e., } \vec{a} = \ddot{r}\hat{i} + (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{j} + \dot{r}\dot{\theta}\hat{j} + r\dot{\theta}^2(-\hat{i})$$

$$\text{i.e., } \vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{i} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{j}$$

Thus radial and transverse acceleration components are

$$\ddot{r} - r\dot{\theta}^2 \text{ and } 2\dot{r}\dot{\theta} + r\ddot{\theta}, \text{ i.e., } \ddot{r} - r\dot{\theta}^2 \text{ and } \frac{1}{r}\frac{d}{dt}(r^2\dot{\theta}).$$

$$\text{Also we have } a^2 = (\ddot{r} - r\dot{\theta}^2)^2 + (2\dot{r}\dot{\theta} + r\ddot{\theta})^2 \text{ and } \tan \beta = \frac{2\dot{r}\dot{\theta} + r\ddot{\theta}}{\ddot{r} - r\dot{\theta}^2},$$

where $a = |\vec{a}|$ and β is the angle which the direction of $\vec{a}$ makes with the radial direction.



Tangential and Normal Components (Intrinsic Components)

Two perpendicular direction in intrinsic systems are tangential direction, i.e., arc length increasing direction and normal direction.

Let a particle be describing a plane curve and at any time t $P(s, \psi)$ be position of it. Let $\hat{i}$ and $\hat{j}$ be the unit vectors along tangential and normal directions. The two perpendicular directions change when the particle moves to a neighboring point Q , i.e., those changes with ψ .

Hence, we have $\frac{d\hat{i}}{d\psi} = \hat{j}$ and $\frac{d\hat{j}}{d\psi} = -\hat{i}$.



Now, the linear velocity at P will be the rate of change of arc length w.r.t.

time and it is $\dot{s} = \frac{ds}{dt}$ along the tangent.

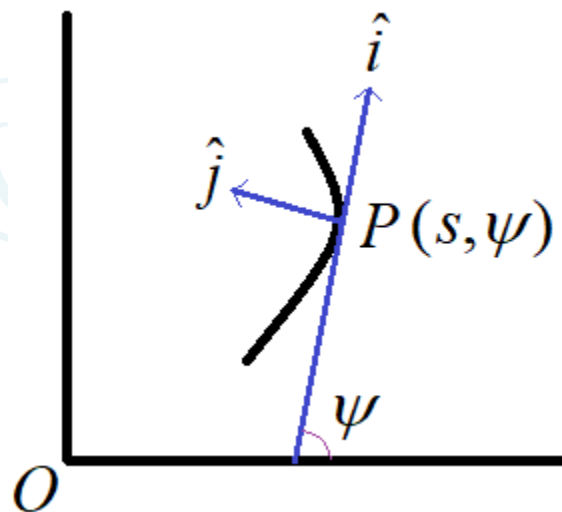
No normal velocity component is present.

Thus, the velocity (linear velocity) $\vec{v}$ is at P may be written as

$$\vec{v} = \dot{s}\hat{i}, \text{ i.e., } |\vec{v}| = v = \dot{s}.$$

So, the acceleration of the particle will be

$$\vec{a} = \frac{d\vec{v}}{dt} = \ddot{s}\hat{i} + \dot{s}\frac{d\hat{i}}{dt} = \ddot{s}\hat{i} + \dot{s}\frac{d\hat{i}}{d\psi}\frac{d\psi}{ds}\frac{ds}{dt}$$



$$\text{i.e., } \vec{a} = \ddot{s}\hat{i} + \dot{s}\hat{j}\frac{1}{\rho}\dot{s} = \ddot{s}\hat{i} + \frac{v^2}{\rho}\hat{j},$$

where $\rho = \frac{ds}{d\psi}$ is the radius of curvature of the curve at P and it may be obtained from intrinsic equation $s = f(\psi)$ of the curve.

Thus, the tangential and normal accelerations are $\ddot{s}$ and $\frac{v^2}{\rho}$.

If the acceleration makes angle α with the tangential direction and $|\vec{a}| = a$

$$\text{then } a^2 = \ddot{s}^2 + \left(\frac{v^2}{\rho}\right)^2 \text{ and } \tan \alpha = \frac{v^2/\rho}{\ddot{s}}.$$



Example: If the radial and transverse velocities of a point are always proportional to each other and this holds for acceleration also, then prove that its velocity will vary as some power of the radius vector.

Solution: It is given that

$$\dot{r} \propto r\dot{\theta} \text{ and } \ddot{r} - r\dot{\theta}^2 \propto \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta})$$

$$\Rightarrow \dot{r} = \lambda r\dot{\theta} \dots\dots (1) \quad \text{and} \quad \ddot{r} - r\dot{\theta}^2 = \frac{\mu}{r} \frac{d}{dt}(r^2\dot{\theta})$$

$$\ddot{r} - r\dot{\theta}^2 = \mu(2\dot{r}\dot{\theta} + r\ddot{\theta}) \dots\dots (2)$$



Let v be the velocity. Then $v = \sqrt{\dot{r}^2 + (r\dot{\theta})^2} = \sqrt{\dot{r}^2 + \frac{\dot{r}^2}{\lambda^2}} \left[\text{as } r\dot{\theta} = \frac{\dot{r}}{\lambda} \right],$

$$\text{i.e., } v = \left(1 + \frac{1}{\lambda^2} \right)^{1/2} \dot{r}.$$

Now from (2) we have

$$\ddot{r} - r \left(\frac{\dot{r}}{\lambda r} \right)^2 = \frac{\mu}{r} \frac{d}{dt} \left(r^2 \frac{\dot{r}}{\lambda r} \right) \Rightarrow \ddot{r} - \frac{\dot{r}^2}{\lambda^2 r} = \frac{\mu}{\lambda r} \frac{d}{dt} (r\dot{r}) = \frac{\mu}{\lambda r} (r\ddot{r} + \dot{r}^2)$$

$$\Rightarrow \ddot{r} - \frac{\dot{r}^2}{\lambda^2 r} = \frac{\mu}{\lambda} \ddot{r} + \frac{\mu \dot{r}^2}{\lambda r} \Rightarrow \ddot{r} - \frac{\mu}{\lambda} \ddot{r} = \frac{\dot{r}^2}{\lambda^2 r} + \frac{\mu r^2}{\lambda r}$$



$$\Rightarrow \ddot{r} \left(1 - \frac{\mu}{\lambda} \right) = \left(\frac{1}{\lambda^2} + \frac{\mu}{\lambda} \right) \frac{\dot{r}^2}{r}$$

$$\Rightarrow A\ddot{r} = B \frac{\dot{r}^2}{r}, \text{ where } A = 1 - \frac{\mu}{\lambda}, B = \frac{1}{\lambda^2} + \frac{\mu}{\lambda}$$

$$\Rightarrow \ddot{r} = C \frac{\dot{r}^2}{r} \text{ (where } C = \frac{B}{A} \text{)} \Rightarrow \frac{\ddot{r}}{\dot{r}} = C \frac{\dot{r}}{r}$$

Integrating we have $\log \dot{r} = C \log r + \log D \Rightarrow \dot{r} = Dr^C$

$$\text{Therefore, } v = \left(1 + \frac{1}{\lambda^2} \right)^{1/2} Dr^C.$$

Thus, the velocity is proportional to any arbitrary power of r .



Example: The velocities of a particle along and perpendicular to radius vector from a fixed origin are λr^2 and $\mu\theta^2$. Show that the equation to the path is $\frac{\lambda}{\theta} = \frac{\mu}{2r} + c$, and the components of acceleration are

$$2\lambda^2 r^3 - \mu^2 \frac{\theta^4}{r} \quad \text{and} \quad \lambda\mu r\theta^2 + 2\mu^2 \frac{\theta^3}{r}.$$

Solution: It is given that

$$\dot{r} = \lambda r^2 \dots\dots (1) \quad \text{and} \quad r\dot{\theta} = \mu\theta^2 \dots\dots (2)$$

Dividing (1) by (2) we have $\frac{\frac{dr}{dt}}{r \frac{d\theta}{dt}} = \frac{\lambda r^2}{\mu\theta^2}$



$$\Rightarrow \mu \frac{dr}{r^3} = \lambda \frac{d\theta}{\theta^2} \Rightarrow -\frac{\mu}{2r^2} = -\frac{\lambda}{\theta} + c \quad (\text{where } c \text{ is constant})$$

$$\Rightarrow \frac{\lambda}{\theta} = \frac{\mu}{2r^2} + c$$

We have

Radial acceleration $= \ddot{r} - r\dot{\theta}^2$ and Transverse acceleration $= 2\dot{r}\dot{\theta} + r\ddot{\theta}$.

From (2) we have $r\dot{\theta} = \mu\theta^2$

Differentiating w.r.t. t we have $\dot{r}\dot{\theta} + r\ddot{\theta} = 2\mu\theta\dot{\theta}$.



Therefore $2\dot{r}\dot{\theta} + r\ddot{\theta} = 2\mu\theta\dot{\theta} + \dot{r}\dot{\theta} = 2\mu\theta\frac{\mu\theta^2}{r} + \lambda r^2\frac{\mu\theta^2}{r} = \lambda\mu r\theta^2 + 2\mu^2\frac{\theta^3}{r}$,

which is the transverse acceleration.

From (1) we have

$$\dot{r} = \lambda r^2 \Rightarrow \ddot{r} = 2\lambda r\dot{r} = 2\lambda r\lambda r^2 \Rightarrow \ddot{r} = 2\lambda^2 r^3.$$

Therefore $\ddot{r} - r\dot{\theta}^2 = 2\lambda^2 r^3 - r\left(\frac{\mu\theta^2}{r}\right) = 2\lambda^2 r^3 - \mu^2\frac{\theta^4}{r}$, which is the radial acceleration.



Example: Prove that the path of a point which possesses two constant velocities one along a fixed direction and the other perpendicular to the radius vector drawn from a fixed point is a conic.

Example: A particle moves along circle $r = 2a \cos \theta$ in such a way that its acceleration towards the origin is always zero. Prove that

$$\frac{d^2\theta}{dt^2} = -2 \cot \theta \dot{\theta}^2.$$



Angular Velocity and Acceleration

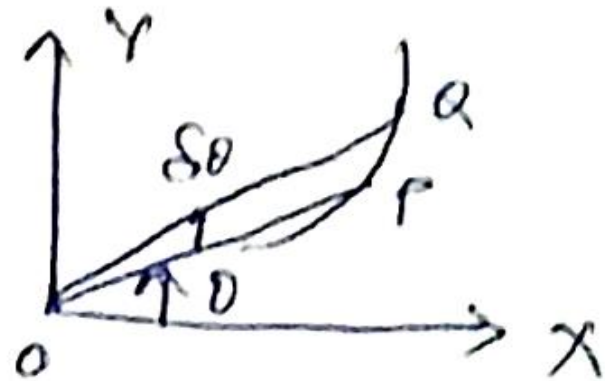
The angular velocity of a point P about another point O is the rate of change of the angle which it makes with some fixed direction.

The angular velocity of P about O

$$= \lim_{\delta t \rightarrow 0} \frac{\delta \theta}{\delta t} = \frac{d\theta}{dt} = \dot{\theta}.$$

Similar, angular acceleration is

$$\frac{d}{dt} \left(\frac{d\theta}{dt} \right) = \frac{d^2\theta}{dt^2} = \ddot{\theta}.$$



Rotation between angular and linear velocities

Let v be the velocity of the particle at $P(r, \theta)$ along the tangent.

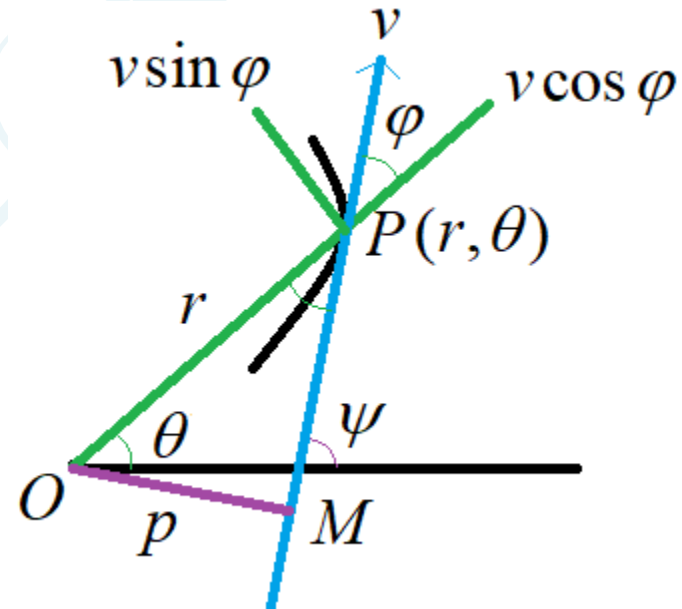
So we have

$$\dot{r} = v \cos \varphi \dots\dots (1)$$

$$r\dot{\theta} = v \sin \varphi \dots\dots (2)$$

We also have from $\triangle OPM$

$$\sin \varphi = \frac{p}{r} \dots\dots (3)$$



From (2) and (3) we get

$$\dot{\theta} = \frac{v \sin \varphi}{r} = \frac{vp}{r^2}.$$

For circular motion it should be (with centre as pole)

$$\dot{\theta} = \frac{v}{a},$$

where a is the radius of the circle.

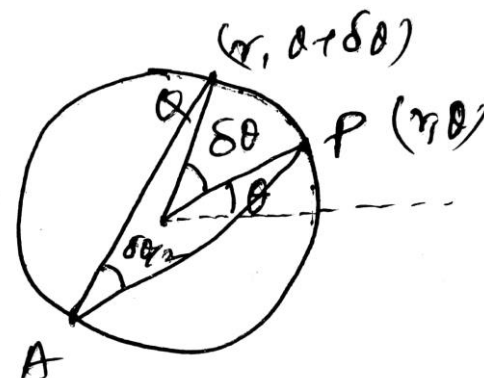


Example: If a point moves along a circle, prove that its angular velocity about any point on the circle is half of that about the centre.

Solution:

Let the angle of any point P be θ and that of a neighbouring point Q be $\theta + \delta\theta$.

Therefore, the change in angle due to the movement of the point from P to Q be $\delta\theta$ with respect to the centre.



Let A be any point on the circle. Then the change in angle of the point due to the movement from P to Q with respect to A is $\frac{\delta\theta}{2}$.

Now, the angular velocity about the centre is $\lim_{\delta t \rightarrow 0} \frac{\delta\theta}{\delta t} = \dot{\theta}$ and that of about

A any point on the circle is $\lim_{\delta t \rightarrow 0} \frac{1}{2} \frac{\delta\theta}{\delta t} = \frac{1}{2} \dot{\theta}$.

Example: A point describes a circle of radius a with a uniform speed v ;

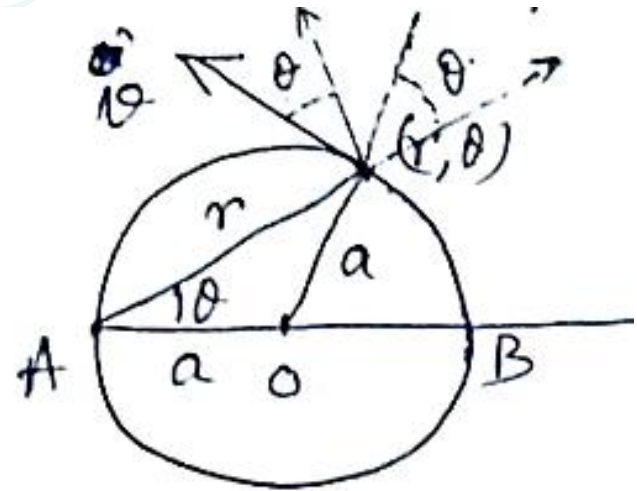
show that the radial and transverse acceleration are $-\frac{v^2}{a} \cos \theta$ and



$-\frac{v^2}{a} \sin \theta$ if a diameter is taken as initial line and end of the diameter as pole.

Solution: The equation of the circle whose or diameter is initial line and one end of that diameter is pole, is given by $r = 2a \cos \theta$ (1)

It is given that a point is moving on circle given by (1) with a uniform speed v .



Therefore the radial velocity $\dot{r} = -v \sin \theta$

and transverse velocity $r\dot{\theta} = v \cos \theta$

$$\therefore \dot{r} = -v \cos \theta \dot{\theta} \text{ and } \dot{\theta} = \frac{v \cos \theta}{r} = \frac{v}{2a}$$

Now, we know that radial acceleration $= \ddot{r} - r\dot{\theta}^2$

and transverse acceleration $= 2\dot{r}\dot{\theta} + r\ddot{\theta}$

$$\text{So, radial acceleration} = -v \cos \theta \frac{v}{2a} - \left(\frac{v}{2a} \right)^2 2a \cos \theta$$

$$= -\frac{v^2}{2a} \cos \theta - \frac{v^2}{2a} \cos \theta = -\frac{v^2}{a} \cos \theta$$



$$\begin{aligned}\text{and transverse acceleration} &= 2\dot{r}\dot{\theta} + r\ddot{\theta} = -2v\sin\theta \frac{v}{2a} + 0 \quad [\text{as } \ddot{\theta}=0] \\ &= -\frac{v^2}{a}\sin\theta.\end{aligned}$$

Example: A point describes uniformly a given straight line, show that its angular velocity about a fixed point varies inversely as the square of its distance from the fixed point.



Example: A point P describes a curve with a constant velocity and the radius vector joining P to a fixed point O has an angular velocity which is inversely proportional to OP ; show that the curve is an equiangular spiral with O as pole and that the acceleration of P along the normal varies inversely as OP .

Example: If the velocity of a point moving in a plane curve varies as the radius of curvature, show that the direction of motion revolves with constant velocity. Also if the angular velocity of the moving point about a fixed origin be constant, show that if transverse acceleration varies as its radial velocity.



Example: A point moves in plane curve so that its tangential acceleration is constant and the magnitudes of tangential velocity and the normal acceleration are in a constant ratio. Show that the intrinsic equation of the path is of the form $s = A\psi^2 + B\psi + C$.

Solution: We have $\ddot{s} = k_1 \dots\dots\dots (1)$

and $\frac{\dot{s}}{v^2 / \rho} = k_2 \Rightarrow \frac{v\rho}{v^2} = k_2 \Rightarrow \rho = k_2 v \dots\dots (2)$

From (1) we have $\ddot{s} = \frac{dv}{dt} = v \frac{dv}{ds} = k_1$
 $\Rightarrow v dv = k_1 ds$



Integrating we get

$$\frac{v^2}{2} = k_1 s + k_3 \Rightarrow v = \sqrt{2k_1 s + k_3}$$

Now from (2) we have

$$\frac{ds}{d\psi} = k_2 \sqrt{2k_1 s + k_3} \Rightarrow \frac{ds}{\sqrt{2k_1 s + k_3}} = k_2 d\psi$$

Integrating we have

$$\begin{aligned} \frac{\sqrt{2k_1 s + k_3}}{2k_1 \cdot \frac{1}{2}} &= k_2 \psi + k_4 \\ \Rightarrow \sqrt{2k_1 s + k_3} &= k_1 k_2 \psi + k_1 k_4 \end{aligned}$$



$$\Rightarrow 2k_1s + k_3 = k_1^2k_2^2\psi^2 + 2k_1^2k_2k_4\psi + k_1^2k_4^2$$

$$\Rightarrow s = \frac{k_1k_2^2}{2}\psi^2 + 2k_1k_2k_4\psi + \frac{k_1^2k_4^2 - k_3}{2k_1}$$

Thus $s = A\psi^2 + B\psi + C$ is the intrinsic equation of the path. (Hence the result)

Example: A point moves in a curve so that its tangential and normal accelerations are equal and the tangent rotates with constant angular velocity. Show that the intrinsic equation of the path is of the form

$$s = Ae^{\psi} + B.$$



Solution: We have $\ddot{s} = \frac{v^2}{\rho}$ (1) and $\frac{d\psi}{dt} = k_1 = \text{constant}$ (2)

From (1) we get $v \frac{dv}{ds} = \frac{v^2}{\rho} \Rightarrow \frac{dv}{ds} = v \frac{d\psi}{ds} \Rightarrow \frac{dv}{v} = d\psi$

Integrating, we obtain

$\log v = \psi + \log k_2$ (k_2 is a constant)

$$\Rightarrow v = k_2 e^{\psi} \Rightarrow \frac{ds}{dt} = \frac{ds}{d\psi} \frac{d\psi}{dt} = k_2 e^{\psi}$$

$$\Rightarrow \frac{ds}{d\psi} k_1 = k_2 e^{\psi} \Rightarrow ds = \frac{k_2}{k_1} e^{\psi} d\psi$$



Integrating we get

$$s = \frac{k_2}{k_1} e^\psi + k_3 \quad (k_3 \text{ is a constant}).$$

Hence the intrinsic equation of the path is of the form $s = Ae^\psi + B$, where A and B are constants.

Example: A particle describes a curve (for which s and ψ vanish simultaneously) with uniform speed v . If the acceleration at any point s be

$\frac{v^2 c}{s^2 + c^2}$, prove that curve is a catenary.



Solution: Since v is constant, then $\ddot{s} = 0$

Then the acceleration given by $\sqrt{(\ddot{s})^2 + \left(\frac{v^2}{\rho}\right)^2} = \frac{v^2}{\rho}$

Therefore, we have

$$\frac{v^2}{\rho} = \frac{v^2 c}{s^2 + c^2} \Rightarrow \frac{d\psi}{ds} = \frac{c}{s^2 + c^2} \Rightarrow d\psi = \frac{c ds}{s^2 + c^2}$$

Integrating, we get

$$\psi = c \frac{1}{c} \tan^{-1}\left(\frac{s}{c}\right) + k \quad \text{i.e., } \psi = \tan^{-1}\left(\frac{s}{c}\right) + k.$$

Now, at $s = 0, \psi = 0 \Rightarrow k = 0$. So, $s = c \tan \psi$



Thus, $\psi = \tan^{-1}\left(\frac{s}{c}\right)$ is the equation of path, which is a catenary. (Proved)

Example: A small bead slides with constant speed v on a smooth wire in the shape of a cardioid $r = a(1 + \cos \theta)$. Show that the value of $\dot{\theta}$ is $\left(v \sec \frac{\theta}{2}\right) / 2a$ and that the radial component of the acceleration is constant.



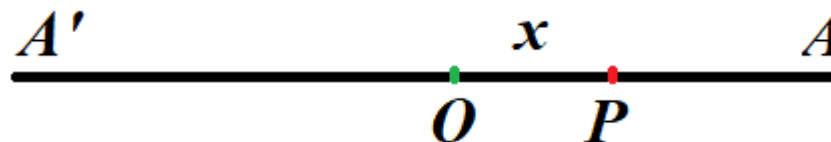
Part – IV

Simple Harmonic Motion



A particle is said to execute Simple Harmonic motion if it moves in a straight line such that its acceleration is always directed towards a fixed point on it and is proportional to the distance from the fixed point.

Let O be a fixed point on the line $A'O A$ and let P be the position of the particle at any time t such that $OP = x$ and the velocity at P be v in the direction OA .



So, the acceleration at P is μx , where μ is a positive constant.



Since the acceleration is always directed towards a fixed point O , i.e. the acceleration is in the direction opposite to that in which x increases, so, the equation of motion of the particle is

$$\frac{d^2x}{dt^2} = -\mu x \quad (1)$$

$$\text{Or, } v \frac{dv}{dx} = -\mu x$$

$$\text{Or, } v dv = -\mu x dx$$

Integrating we get

$$\frac{v^2}{2} = -\frac{\mu x^2}{2} + c_1 \quad (2)$$



Since the particle is moving in the direction OA and the acceleration is given to be directed towards the fixed point O , i.e., in the opposite direction, so the particle must come into rest at some point on the line.

Let the point be A such that $OA = a$.

Then putting at $x = a, v = 0$ in (2), we get $c_1 = \frac{\mu a^2}{2}$

Hence from (2) we have

$$v^2 = \mu(a^2 - x^2) \text{ Or, } v = \pm \sqrt{\mu} \sqrt{(a^2 - x^2)} \quad (3)$$

This is the expression of the velocity of the particle at P for any displacement x from O .



Case I: Let the time t be measured from the instant when the particle is at O .

In this case, as t increases x also increases and hence $\frac{dx}{dt}$ is positive.

Therefore, from (3), we get

$$v = \sqrt{\mu} \sqrt{(a^2 - x^2)} \quad \text{Or,} \quad \frac{dx}{dt} = \sqrt{\mu} \sqrt{(a^2 - x^2)} \quad \text{Or,} \quad \frac{dx}{\sqrt{(a^2 - x^2)}} = \sqrt{\mu} dt$$

$$\text{Integrating, we get } \sin^{-1}\left(\frac{x}{a}\right) = \sqrt{\mu} t + c_2, \quad (4)$$



where c_2 is arbitrary constant.

$$\text{Or } x = a \sin(\sqrt{\mu}t + c_2)$$

Initially, $x = 0$ at $t = 0$. So, using this in (4) we get $c_2 = 0$

$$\text{Hence, we have } x = a \sin(\sqrt{\mu}t) \quad (5)$$

Thus (5) gives the position of the particle at any time t , when t is measured from the fixed point O .

As the particle proceeds towards A and the acceleration being towards O , the velocity goes on decreasing as x increases and hence it must vanish at A such that $OA = a$. So the particle is at rest for an instant at A . Then since



the acceleration is directed towards O , the particle moves in the direction AO with a velocity which increases as x decrease and it is maximum at O . Due to this velocity the particle proceeds further to the opposite side A , i.e., towards A' . The velocity again goes on decreasing since the acceleration is directed towards O and the particle comes to rest at A' . Again it moves towards O and thus it is continuously moving between A and A' .

The nature of the motion is therefore oscillatory.



Case II: Let the time t be measured from the instant when the particle is at A (the extreme point or position of rest).

Since x is always measured from the fixed point O and in this case the particle moves towards O , i.e. in the direction AO , so, x decreases as t increases. Therefore, $\frac{dx}{dt}$ is negative.

Hence from (3) we get $\frac{dx}{dt} = -\sqrt{\mu}\sqrt{(a^2 - x^2)}$ Or, $-\frac{dx}{\sqrt{(a^2 - x^2)}} = \sqrt{\mu}dt$

Integrating we get $\cos^{-1}\left(\frac{x}{a}\right) = \sqrt{\mu}t + c_3$ (6)



where c_3 is an arbitrary constant.

Initially at A, $x = a, t = 0$. Using this in (6) we get $c_3 = 0$

Hence from (6) we have $x = a \cos(\sqrt{\mu}t)$ (7)

Thus (7) gives the position of the particle at any time t , when t is measured from the extreme position.



Amplitude:

The maximum displacement of the particle on either side of the centre of oscillation is called the amplitude of the motion.

Here the amplitude of the simple harmonic motion(SHM) is a .

Periodic Motion:

A particle is said to have periodic motion if it moves in such a manner that it occupies the same position after a certain fixed interval of time and moves in the same direction with same velocity.



In SHM, we have

$$x = a \cos(\sqrt{\mu}t), \quad (i)$$

where t is measured from the position of rest.

$$\text{Or, } x = a \cos(2\pi + \sqrt{\mu}t) = a \cos\left\{\sqrt{\mu}\left(t + \frac{2\pi}{\sqrt{\mu}}\right)\right\}$$

where a is the amplitude of the motion and x is measured from the mean position.

Now, differentiating both sides of (i) w.r.t t we have

$$\frac{dx}{dt} = -a\sqrt{\mu} \sin(\sqrt{\mu}t), \text{ Or, } v = -a\sqrt{\mu} \sin(2\pi + \sqrt{\mu}t)$$



$$\text{So, } v = -a\sqrt{\mu} \sin \left\{ \sqrt{\mu} \left(t + \frac{2\pi}{\sqrt{\mu}} \right) \right\}$$

Thus, the values of x and v are repeated when t increased by $\frac{2\pi}{\sqrt{\mu}}$.

Hence after every interval of $\frac{2\pi}{\sqrt{\mu}}$, we have the same position and same velocity in the same direction and consequently, SHM is periodic, the period being $\frac{2\pi}{\sqrt{\mu}}$, which is independent of the amplitude a .



Period:

The time required for a complete oscillation is called the periodic time or the period of the oscillation or simply period.

Frequency:

The number of complete oscillations in one second is called frequency.

Thus if n be the frequency and T be the periodic time, then $nT = 1$ or

$$n = \frac{1}{T}. \text{ So for S.H.M the frequency } n = \frac{\sqrt{\mu}}{2\pi}$$



Phase and Epoch:

Re-writing the equation of SHM we have $\frac{d^2x}{dt^2} + \mu x = 0$ which is a linear equation with constant coefficient and its general solution is

$$x = a \cos(\sqrt{\mu}t + \varepsilon) \quad (\text{ii})$$

The constant ε is called the starting phase or the epoch of the motion and the quantity $(\sqrt{\mu}t + \varepsilon)$ called the argument of the motion.

The phase at any time t of a S.H.M is the time that has elapsed since the particle was at its maximum distance in the positive direction.



From (ii), we get x is maximum when $\cos(\sqrt{\mu}t + \varepsilon)$ is maximum, i.e.

$$\text{when } \cos(\sqrt{\mu}t + \varepsilon) = 1$$

Clearly, if x is maximum at time t_0 , then $\cos(\sqrt{\mu}t_0 + \varepsilon) = 1$

$$\text{So, } \sqrt{\mu}t_0 + \varepsilon = 0 \Rightarrow t_0 = -\frac{\varepsilon}{\sqrt{\mu}}.$$

$$\text{Phase at } t = t - t_0 = t + \frac{\varepsilon}{\sqrt{\mu}} = \frac{\sqrt{\mu}t + \varepsilon}{\sqrt{\mu}} = \frac{\text{argument}}{\sqrt{\mu}}$$



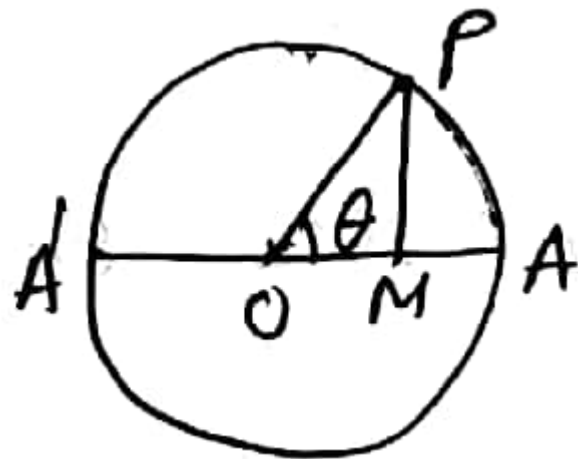
A geometrical representation of the S.H.M:

Let a particle P move on a circle with constant angular velocity ω and let M be the foot of the perpendicular from P on any diameter $A'O A$. If a be the radius of the circle, the only acceleration of P is $\omega^2 a$ towards O .

If $\angle AOP = \theta$ and $OM = x$, the component of this acceleration along OA is $\omega^2 a \cos \theta = \omega^2 a \frac{x}{a} = \omega^2 x$ towards O .

Hence the equation of the motion of the point M is $\ddot{x} = -\omega^2 x$

This is a S.H.M.



Thus, if a particle describes a circle with constant angular velocity the foot of the perpendicular from it on any diameter executes a simple harmonic motion.

Note 1: Acceleration of a particle at a distance x from the fixed point O is μx .

Note 2: The acceleration is maximum at the extreme position, i.e. when $x = a$ and thus $f_{\max} = \mu a$

Note 3: The velocity is maximum at the mean position, i.e., when $x = 0$.

Thus $v_{\max} = \sqrt{\mu a}$



Time period in terms of velocities and accelerations at two positions:

Let accelerations and velocities at two positions x_1 and x_2 from the centre be a_1, a_2 and v_1, v_2 respectively.

Thus, we have $a_1 = -\mu x_1, a_2 = -\mu x_2$

$$\text{and } v_1^2 = \mu(a^2 - x_1^2), v_2^2 = \mu(a^2 - x_2^2)$$

Then we can write

$$a_1 - a_2 = \mu(x_2 - x_1), v_1^2 - v_2^2 = \mu(x_2^2 - x_1^2) \text{ and } a_2^2 - a_1^2 = \mu^2(x_2^2 - x_1^2)$$

$$\text{Therefore, } \mu = \frac{a_2^2 - a_1^2}{v_1^2 - v_2^2}.$$



Hence the time period T can be written as $T = 2\pi \sqrt{\frac{v_1^2 - v_2^2}{a_2^2 - a_1^2}}$.

We also have $T = 2\pi \sqrt{\frac{x_2 - x_1}{a_1 - a_2}}$ and $T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$.

Amplitude in terms of velocities and acceleration at two positions:

From above we have $a^2 = \frac{1}{\mu}(v_1^2 + \mu x_1^2)$

$$\Rightarrow a^2 = \frac{1}{\mu} \left(v_1^2 + \frac{a_1^2}{\mu} \right) = \frac{1}{\mu^2} (v_1^2 \mu + a_1^2), \quad \left[\text{as } x_1^2 = \frac{a_1^2}{\mu^2} \right]$$



$$\begin{aligned}
 \text{Then } a^2 &= \frac{(v_1^2 - v_2^2)^2}{(a_2^2 - a_1^2)^2} \left(v_1^2 \frac{(a_2^2 - a_1^2)}{v_1^2 - v_2^2} + a_1^2 \right) \\
 &= \frac{(v_1^2 - v_2^2)}{(a_2^2 - a_1^2)^2} (v_1^2 a_2^2 - v_1^2 a_1^2 + v_1^2 a_1^2 - v_2^2 a_1^2) \\
 &= \frac{(v_1^2 - v_2^2)}{(a_2^2 - a_1^2)^2} (v_1^2 a_2^2 - v_2^2 a_1^2)
 \end{aligned}$$

$$\text{Therefore, } a = \frac{1}{(a_2^2 - a_1^2)} \left\{ (v_1^2 - v_2^2) (v_1^2 a_2^2 - v_2^2 a_1^2) \right\}^{1/2}.$$



Velocity and position in terms of time period:

We have $v^2 = \mu(a^2 - x^2)$ and $x = a \cos \sqrt{\mu}t$ with $\sqrt{\mu} = \frac{2\pi}{T}$

$$\therefore v^2 = \frac{4\pi^2}{T^2}(a^2 - x^2) \text{ and } x = a \cos \frac{2\pi}{T}t$$

Thus the maximum velocity $v_{\max} = \frac{2\pi}{T}a$.



Time between two positions of SHM:

Let at times t_1 and t_2 the particles at position x_1 and x_2 respectively.

$$\text{We have } v = -\sqrt{\mu}\sqrt{a^2 - x^2} \text{ i.e., } \frac{dx}{dt} = -\sqrt{\mu}\sqrt{a^2 - x^2}$$

$$\text{i.e., } \sqrt{\mu}dt = -\frac{dx}{\sqrt{a^2 - x^2}}$$

Now integrating the above differential equation from t_1 to t_2 in the left side and from x_1 to x_2 in the right side we get

$$\sqrt{\mu} \int_{t_1}^{t_2} dt = \int_{x_1}^{x_2} -\frac{dx}{\sqrt{a^2 - x^2}}$$



$$\Rightarrow t_2 - t_1 = \frac{1}{\sqrt{\mu}} \left(\cos^{-1} \frac{x_2}{a} - \cos^{-1} \frac{x_1}{a} \right)$$

If the time is t^* moves from $x = b$ to the end point in the same side (while moving towards the end)

$$t^* = \frac{1}{\sqrt{\mu}} \cos^{-1} \frac{b}{a} = \frac{1}{\sqrt{\mu}} \sin^{-1} \frac{\sqrt{a^2 - b^2}}{a} = \frac{1}{\sqrt{\mu}} \tan^{-1} \frac{\sqrt{a^2 - b^2}}{b}.$$



Time period and maximum velocity in case positions of two zero velocities are on the same side of the origin:

Let both such positions lie on one side of the origin O at a distance $OA' = a$ and $OA = b > a$ respectively.

Then the centre C of the S.H.M lies between the two at a distance $\frac{a+b}{2}$

from origin and at a distance $\frac{b-a}{2}$ from both positions A and A' .

Therefore the amplitude of the SHM is $\frac{b-a}{2}$.



$$\therefore v_{\max} = \sqrt{\mu} \frac{b-a}{2} \text{ and } T = \frac{\pi(b-a)}{v_{\max}}.$$

The equation of S.H.M will be

$$\left[\frac{d^2}{dt^2} \left(x - \frac{b+a}{2} \right) = -\mu \left(x - \frac{b+a}{2} \right) \right]$$

$$\Rightarrow \frac{d^2 x}{dt^2} = -\mu \left(x - \frac{a+b}{2} \right).$$



Average velocity and acceleration in SHM:

$$\text{Average velocity} = \frac{4}{T} \int_0^{T/4} v dt \quad \text{and} \quad \text{average acceleration} = \frac{4}{T} \int_0^{T/4} f dt$$

where T is period of SHM and $x = a \cos(\sqrt{\mu}t)$, $v = -a\sqrt{\mu} \sin(\sqrt{\mu}t)$,

$$f = -a\mu \cos(\sqrt{\mu}t)$$

Example: A point in a straight line with SHM has velocities v_1 and v_2 when its distances from the centre are x_1 and x_2 . Show that the period of

$$\text{the motion is } 2\pi \frac{\sqrt{x_1^2 - x_2^2}}{v_2^2 - v_1^2}.$$



Example: A point executes SHM such that in two of its positions the velocities are u, v and the corresponding accelerations are α, β show that the distance between the positions is $\frac{v^2 - u^2}{\alpha + \beta}$ and the amplitude of the

motion is $\frac{\left[(v^2 - u^2)(\alpha^2 v^2 - \beta^2 u^2) \right]^{1/2}}{\beta^2 - \alpha^2}$.

Solution: We have $\alpha = -\mu x_1, \beta = -\mu x_2$ where x_1 and x_2 are positions for the accelerations α and β and the equation of SHM is $\frac{d^2 x}{dt^2} = -\mu x$.



$$\text{Also } u^2 = \mu(a^2 - x_1^2), v^2 = \mu(a^2 - x_2^2)$$

$$\therefore v^2 - u^2 = \mu(x_1^2 - x_2^2) \dots\dots\dots (1) \quad \text{and } \alpha + \beta = -\mu(x_1 + x_2) \dots\dots\dots (2)$$

Dividing (1) by (2) we get

$$(x_2 - x_1) = \frac{v^2 - u^2}{\alpha + \beta} \quad (\text{Part 1 proved})$$

$$\text{We also have, } \beta^2 - \alpha^2 = \mu^2(x_2^2 - x_1^2) \dots\dots\dots (3)$$

$$u^2 - v^2 = \mu(x_2^2 - x_1^2) \dots\dots\dots (4)$$

$$\text{So, } \mu = \frac{\beta^2 - \alpha^2}{u^2 - v^2}.$$



From above $a^2 = \frac{u^2}{\mu} + x_1^2 = \frac{u^2}{\mu} + \frac{\alpha^2}{\mu^2} = \frac{1}{\mu^2}(\mu u^2 + \alpha^2)$

•

•

•



Example: Show that in a SHM the average speed and the average acceleration (in magnitude) are obtained by multiplying their maximum values by 0.637.

Solution: Let the position of the particle executing SHM be $x = a \cos(\sqrt{\mu}t)$, where a is the amplitude of the SHM, μ is constant and the time period be T . Then $T = \frac{2\pi}{\sqrt{\mu}}$. Let v and f be velocity and acceleration at time t .

We have average velocity $= \frac{4}{T} \int_0^{T/4} v dt$



$$\begin{aligned}
 &= \frac{4}{T} \int_0^{T/4} \left\{ -a\sqrt{\mu} \sin(\sqrt{\mu}t) \right\} dt = \frac{4a}{T} \left[-\cos(\sqrt{\mu}t) \right]_0^{T/4} = \frac{4a}{T} \left[-\cos\left(\sqrt{\mu} \frac{T}{4}\right) \right] \\
 &= -\frac{4a}{T}.
 \end{aligned}$$

$$\text{Average acceleration} = \frac{4}{T} \int_0^{T/4} f dt$$

$$\begin{aligned}
 &= \frac{4}{T} \int_0^{T/4} \left\{ -a\mu \cos(\sqrt{\mu}t) \right\} dt = -\frac{4a\sqrt{\mu}}{T} \left[\sin(\sqrt{\mu}t) \right]_0^{T/4} \\
 &= -\frac{4a\sqrt{\mu}}{T} \sin\left(\sqrt{\mu} \frac{T}{4}\right) = -\frac{4a \frac{2\pi}{T}}{T} \sin\left(\frac{2\pi}{T} \frac{T}{4}\right) = -\frac{4a2\pi}{T^2} = -\frac{8a\pi}{T^2}.
 \end{aligned}$$



We know, $v_{\max} = \sqrt{\mu a} = \frac{2\pi}{T} a$ and $f_{\max} = \mu a = \frac{4\pi^2}{T^2} a$.

So, $\frac{v_{ave}}{v_{\max}} = \frac{2}{\pi} = 0.637$ (approx.) and $\frac{f_{ave}}{f_{\max}} = \frac{2}{\pi} = 0.637$.

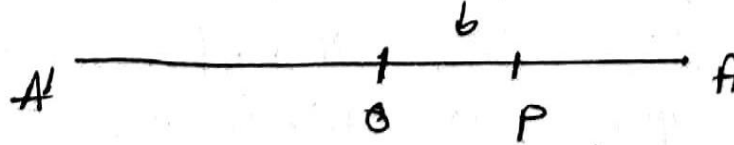
Example: A body moving in straight line OAB with S.H.M has zero velocity when at points A and B whose distance from O are a and b respectively and has a velocity v when half-way between them show that the complete period is $\frac{\pi(b-a)}{v}$.



Example: A particle is performing SHM of period T about a centre O and it passes through a point P ($OP=b$) with velocity v in the direction OP .

Prove that the time which elapses before it returns to P is $\frac{T}{\pi} \tan^{-1} \frac{vT}{2\pi b}$

Solution:



$$v = \frac{dx}{dt} = -\sqrt{\mu} \sqrt{a^2 - x^2} \dots\dots\dots (1)$$

$$\Rightarrow -\frac{dx}{\sqrt{a^2 - x^2}} = \sqrt{\mu} dt$$



Integrating from $x = b$ to $x = a$ we have

$$\int_b^a -\frac{dx}{\sqrt{a^2 - x^2}} = \sqrt{\mu} \int_{t_1}^{t_2} dt,$$

where t_1 is the time at $x = b$ and t_2 is the time at $x = a$

$$\text{So, } (t_2 - t_1) = \frac{1}{\sqrt{\mu}} \cos^{-1} \frac{b}{a} = \frac{1}{\sqrt{\mu}} \tan^{-1} \frac{\sqrt{a^2 - b^2}}{b}$$

From (1) we have for $x = b$ $v = -\sqrt{\mu} \sqrt{a^2 - b^2}$.

Also we have $\sqrt{\mu} = \frac{2\pi}{T}$ as T is the period.



$$\therefore (t_2 - t_1) = \frac{T}{2\pi} \tan^{-1} \left(-\frac{vT}{2\pi b} \right)$$

So the time from $x = b$ to $x = a$ is

$$\frac{T}{2\pi} \tan^{-1} \left(\frac{vT}{2\pi b} \right).$$

Example: A particle is moving with SHM and while making an excursion from one position of rest to the other its distance from the middle point of



its path at three consecutive seconds are observed to be x_1, x_2, x_3 . Prove that the time of a complete revolution is $\frac{2\pi}{\cos^{-1}\left(\frac{x_1 + x_3}{2x_2}\right)}$

Example: A particle starts from rest under an acceleration k^2x directed towards a fixed point and after time t another particle starts from the same position under the same acceleration. Show that the particles will collide at time $\frac{\pi}{k} + \frac{t}{2}$ after the start of the first particle provided $t < \frac{2\pi}{k}$.



Part – V

Central Orbit and Kepler's Laws



Let us suppose that the force F acting on a particle of mass m has the following characteristics –

- (i) It is always directed towards a fixed point.
- (ii) The magnitude of the force F is a force of distance r of the particle from the fixed point.

Such a force is called central force and the fixed point is called the centre of force. The path is described by the particle is called a central orbit.

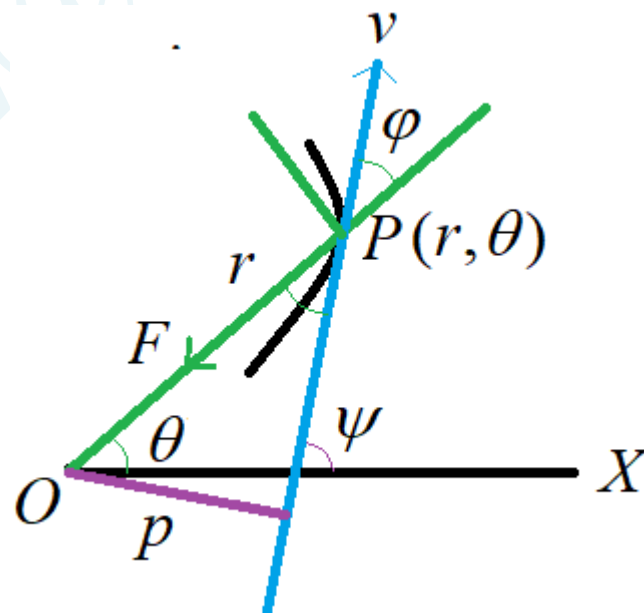


Equation of Motion (for a Central Orbit):

Let O be the centre of force. Let OX be a fixed line through O . Let P be the position of the particle at any time t . With O as pole and OX as initial line let (r, θ) be the coordinates of P .

Let F be the central force per unit mass acting along the line PO .

There is no force acting along the cross radial direction. Hence the equation of



motion of the particle along radial and transverse directions are

$$m \left\{ \frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2 \right\} = -mF, \text{ i.e., } \frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2 = -F, \quad (1)$$

$$m \frac{1}{r} \frac{d}{dt} \left(r^2 \frac{d\theta}{dt} \right) = 0, \text{ i.e., } \frac{d}{dt} \left(r^2 \frac{d\theta}{dt} \right) = 0, \quad (2)$$

where m is the mass of the particle.

Integrating (2) we have

$$r^2 \frac{d\theta}{dt} = \text{constant} = h \text{ (say)} \quad (3)$$



Case-I: Equation in r and θ

We have $\dot{r} = \frac{dr}{d\theta} \cdot \frac{d\theta}{dt} = \frac{h}{r^2} \frac{dr}{d\theta}$

$$\begin{aligned}\ddot{r} &= \frac{d}{dt} \left(\frac{h}{r^2} \frac{dr}{d\theta} \right) = \frac{d}{d\theta} \left(\frac{h}{r^2} \frac{dr}{d\theta} \right) \dot{\theta} \\ &= \frac{h}{r^2} \left[\frac{d}{dr} \left(\frac{h}{r^2} \right) \left(\frac{dr}{d\theta} \right)^2 + \frac{h}{r^2} \frac{d^2 r}{d\theta^2} \right] = -\frac{2h^2}{r^5} \left(\frac{dr}{d\theta} \right)^2 + \frac{h^2}{r^4} \frac{d^2 r}{d\theta^2}\end{aligned}$$

Then from (1) we have

$$\frac{h^2}{r^4} \frac{d^2 r}{d\theta^2} - \frac{2h^2}{r^5} \left(\frac{dr}{d\theta} \right)^2 - r \frac{h^2}{r^4} = -F$$



$$\Rightarrow h^2 r \frac{d^2 r}{d\theta^2} - 2h^2 \left(\frac{dr}{d\theta} \right)^2 - r^2 h^2 = -r^5 F \quad (4)$$

Case-II: Equation in u and θ

Let $u = \frac{1}{r}$, i.e., $r = \frac{1}{u}$,

Then from (3) we get $\dot{\theta} = hu^2$ (5)

$$\text{Now } \dot{r} = \frac{dr}{d\theta} \frac{d\theta}{dt} = \frac{d}{d\theta} \left(\frac{1}{u} \right) \frac{d\theta}{dt} = -\frac{1}{u^2} \frac{du}{d\theta} \frac{d\theta}{dt} = -\frac{1}{u^2} \frac{du}{d\theta} hu^2 = -h \frac{du}{d\theta}$$

$$\ddot{r} = -h \frac{d}{dt} \left(\frac{du}{d\theta} \right) = -h \frac{d}{d\theta} \left(\frac{du}{d\theta} \right) \frac{d\theta}{dt} = -h \frac{d^2 u}{d\theta^2} hu^2 = -h^2 u^2 \frac{d^2 u}{d\theta^2}$$



Then from (1) we get

$$\begin{aligned} -h^2 u^2 \frac{d^2 u}{d\theta^2} - \frac{1}{u} h^2 u^4 &= -F \Rightarrow h^2 u^2 \frac{d^2 u}{d\theta^2} + h^2 u^3 = F \\ \Rightarrow u^2 \left(u + \frac{d^2 u}{d\theta^2} \right) &= \frac{F}{h^2}, \text{ Or, } u + \frac{d^2 u}{d\theta^2} = \frac{F}{h^2 u^2} \end{aligned} \quad (6)$$

This equation of motion of central orbit is used in practical purpose.



Pedal Equation of Central Orbit:

Let the perpendicular from O on the tangent on the orbit at P be ON and let $ON=p$

Then $p = r \sin \phi$ and $\tan \phi = r \frac{d\theta}{dr}$

$$\text{So, } \frac{1}{p^2} = \frac{1}{r^2} \operatorname{cosec}^2 \phi = \frac{1}{r^2} (1 + \cot^2 \phi) = \frac{1}{r^2} \left(1 + \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 \right)$$

$$\text{i.e., } \frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$$



We know $r = \frac{1}{u}$ then $\frac{dr}{d\theta} = -\frac{1}{u^2} \frac{du}{d\theta}$.

Then we have $\frac{1}{p^2} = u^2 + \left(\frac{du}{d\theta}\right)^2$.

Now, differentiating both sides with respect to θ we get

$$-\frac{2}{p^3} \frac{dp}{dr} \frac{dr}{d\theta} = 2u \frac{du}{d\theta} + 2 \frac{du}{d\theta} \frac{d^2u}{d\theta^2}$$
$$\Rightarrow -\frac{2}{p^3} \frac{dp}{dr} \left(-\frac{1}{u^2} \frac{du}{d\theta} \right) = 2 \frac{du}{d\theta} \left(u + \frac{d^2u}{d\theta^2} \right)$$



$$\Rightarrow \frac{1}{p^3} \frac{dp}{dr} = u^2 \left(u + \frac{d^2u}{d\theta^2} \right) \Rightarrow \frac{1}{p^3} \frac{dp}{dr} = \frac{F}{h^2}$$

$$\Rightarrow \frac{h^2}{p^3} \frac{dp}{dr} = F \quad (7)$$

Linear velocity at a position:

We know that the angular velocity of a particle is given by $\dot{\theta} = \frac{vp}{r^2}$

$$\Rightarrow \frac{h}{r^2} = \frac{vp}{r^2} \Rightarrow v = \frac{h}{p} \Rightarrow v \propto \frac{1}{p}. \text{ We also have } v^2 = h^2 \left[u^2 + \left(\frac{du}{d\theta} \right)^2 \right].$$

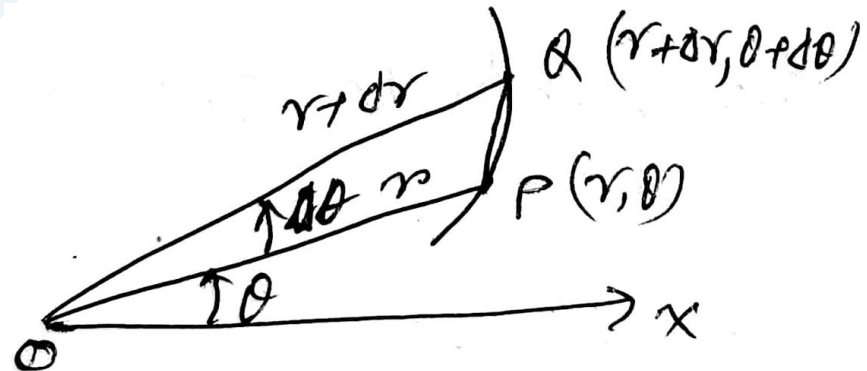


Significance of h :

In a central orbit the sectorial area traced out by the radius vector through the centre of force to the particle per unit time is constant and equal to $\frac{1}{2}h$. (The rate of description of the sectorial area)

Proof: Let $P(r, \theta)$ and $Q(r + \Delta r, \theta + \Delta \theta)$ be the positions of the moving particle on the central orbit at time t and $t + \Delta t$ respectively.

The radius vector OP has described the sectorial area OPQ



in time Δt . As $\Delta t \rightarrow 0, \Delta\theta \rightarrow 0$ and $\Delta r \rightarrow 0$ and arc $PQ \rightarrow$ chord PQ .

Then the rate of description of the sectorial area

$$= \lim_{\Delta t \rightarrow 0} \frac{\text{area of the sector OPQ}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\text{area of the triangle OPQ}}{\Delta t}$$

$$= \lim_{\Delta t \rightarrow 0} \frac{\frac{1}{2} r(r + \Delta r) \sin \Delta\theta}{\Delta t} = \lim_{\Delta t \rightarrow 0} \left(\frac{1}{2} r(r + \Delta r) \frac{\sin \Delta\theta}{\Delta\theta} \frac{\Delta\theta}{\Delta t} \right)$$

$$= \frac{1}{2} \lim_{\Delta t \rightarrow 0} r(r + \Delta r) \lim_{\Delta\theta \rightarrow 0} \frac{\sin \Delta\theta}{\Delta\theta} \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{1}{2} r(r + 0) \cdot 1 \cdot \frac{d\theta}{dt} = \frac{1}{2} r^2 \dot{\theta} = \frac{1}{2} h$$

The rate of describing the sectorial area by the radius vector to the particle moving under a centre force is called areal velocity.



Therefore, $h = 2 \times$ rate of describing the sectorial area
 $= 2 \times$ areal velocity

Angular momentum of the particle is the moment of momentum of the particle about the pole (or origin).

The radial and transverse components of momentum are $m\dot{r}$ and $m r \dot{\theta}$.

Therefore, angular momentum = moment of momentum

$$= r \cdot m r \dot{\theta} = m r^2 \dot{\theta} = m h$$

Thus, $h =$ Angular momentum per unit mass



Hence, the angular momentum of a particle moving under central force is constant.

Some cases when polar equation of the central orbit is not given:

(I) Law of force is given

Suppose $F \propto \frac{1}{r^7} \Rightarrow u + \frac{d^2u}{d\theta^2} \propto u^5$

Integrating, $u^2 + \left(\frac{du}{d\theta}\right)^2 \propto u^6 \Rightarrow v^2 \propto u^6 \Rightarrow v \propto u^3$.

So, $p \propto r^3 \Rightarrow p = ar^3$.



(II) Law of velocity is given

Suppose $v \propto \frac{1}{r^3} \Rightarrow u^2 + \left(\frac{du}{d\theta} \right)^2 \propto u^6$

Differentiate w.r.t. θ

$$\Rightarrow u + \frac{d^2u}{d\theta^2} \propto u^5 \Rightarrow \frac{F}{u^2} \propto u^5 \Rightarrow F \propto u^5 \Rightarrow F \propto \frac{1}{r^7}$$

Also, $p \propto r^3 \Rightarrow p = ar^3$.

(III) Pedal Equation is given

Suppose $a^2 p = r^3 \Rightarrow p \propto r^3 \Rightarrow v \propto u^3$ and so on.



Polar Equation of the central orbit is given:

1) Elliptic Orbit:

Let the central orbit be an ellipse whose equation is

$$\frac{l}{r} = 1 + e \cos \theta,$$

where $l = \frac{b^2}{a}$ is the semi-latus rectum, a is the semi-major axis, b is the semi-minor axis and $b^2 = a^2(1 - e^2)$, i.e., $e^2 = 1 - \frac{b^2}{a^2}$, ($e < 1$) is eccentricity.



We have $u = \frac{1 + e \cos \theta}{l}$.

So, $\frac{du}{d\theta} = -\frac{e \sin \theta}{l}$ and $\frac{d^2u}{d\theta^2} = -\frac{e \cos \theta}{l}$.

From equation of central orbit, we have

$$\frac{F}{h^2 u^2} = \frac{1}{l} \Rightarrow F = \frac{h^2}{l} u^2 = \frac{\mu}{r^2} \text{ where } \mu = \frac{h^2}{l}, \quad \left[u + \frac{d^2u}{d\theta^2} = \frac{F}{h^2 u^2} \right]$$

where h is angular momentum per unit mass.

Thus, the central force for an elliptic varies inversely as the square of radius vector.



We have $v^2 = h^2 \left[u^2 + \left(\frac{du}{d\theta} \right)^2 \right]$

$$v^2 = \mu l \left[\left(\frac{1 + e \cos \theta}{l} \right)^2 + \left(-\frac{e \sin \theta}{l} \right)^2 \right]$$

$$= \frac{\mu l}{l^2} [1 + 2e \cos \theta + e^2 \cos^2 \theta + e^2 \sin^2 \theta]$$

$$= \frac{\mu}{l} [1 + 2e \cos \theta + e^2]$$

$$= \frac{\mu}{l} \left[1 + 2e \cos \theta + 1 - \frac{b^2}{a^2} \right]$$



$$= \mu \left[\frac{2(1 + e \cos \theta)}{l} - \frac{b^2}{a^2 l} \right] = \mu \left[\frac{2}{r} - \frac{1}{a} \right]$$

$\Rightarrow v^2 < \frac{2\mu}{r}$ and the linear velocity varies inversely to square root of radius vector of that position.

$$\text{We have } v^2 = \frac{h^2}{p^2} = \mu \left[\frac{2}{r} - \frac{1}{a} \right]$$

$$\Rightarrow \frac{\mu l}{p^2} = \mu \left[\frac{2}{r} - \frac{1}{a} \right]$$



$$\Rightarrow \frac{b^2/a}{p^2} = \frac{2}{r} - \frac{1}{a} \Rightarrow \frac{b^2}{p^2} = \frac{2a}{r} - 1.$$

This is the pedal equation of the ellipse.

2) Hyperbolic orbit:

$$\frac{l}{r} = 1 + e \cos \theta, e > 1, l = \frac{b^2}{a}, e^2 = 1 + \frac{b^2}{a^2}$$

Similarly, law of force $F \propto \frac{1}{r^2} = \frac{\mu}{r^2}, h^2 = \mu l$



The velocity $v^2 = \mu \left(\frac{2}{r} + \frac{1}{a} \right) > \frac{2\mu}{r}$ and the pedal equation $\frac{b^2}{p^2} = \frac{2a}{r} + 1$.

3) Parabolic orbit:

$\frac{l}{r} = 1 + e \cos \theta$, $e = 1$, $l = 2a$. So the equation becomes $\frac{2a}{r} = 1 + \cos \theta$.

Law of force: $F = \frac{\mu}{r^2}$, with $h^2 = \mu l = 2a\mu$

So, the velocity: $v^2 = \frac{2\mu}{r}$

The pedal equation: $\frac{h^2}{p^2} = \frac{2\mu}{r} \Rightarrow \frac{2a\mu}{p^2} = \frac{2\mu}{r} \Rightarrow p^2 = ar$



So, for $v^2 < \frac{2\mu}{r}$, central orbit is ellipse

$v^2 > \frac{2\mu}{r}$, central orbit is hyperbola

$v^2 = \frac{2\mu}{r}$, central orbit is parabola

where $\mu = \frac{h^2}{l}$.



4) *The curve is $r^n = a^n \cos n\theta$, where a is a parameter of the curve:*

Putting $u = \frac{1}{r}$ i.e. $r = \frac{1}{u}$ we have, $1 = a^n u^n \cos n\theta$

Taking logarithm and then differentiating w.r.t θ we get

$$n \log u + \log \cos n\theta + n \log a = 0$$

$$\frac{n}{u} \frac{du}{d\theta} - n \tan n\theta = 0 \Rightarrow \frac{du}{d\theta} = u \tan n\theta$$

$$\text{So, } \frac{d^2u}{d\theta^2} = u \left(\tan^2 n\theta + n \sec^2 n\theta \right).$$



Then from the equation of central orbit we get

$$F = h^2 u^2 \left(u + \frac{d^2 u}{d\theta^2} \right) = h^2 u^2 \left[u + u \left(\tan^2 n\theta + n \sec^2 n\theta \right) \right]$$

$$= h^2 u^3 \left[1 + \tan^2 n\theta + n \sec^2 n\theta \right]$$

$$= h^2 u^3 (n+1) (a^n u^n)^2$$

$$= h^2 a^{2n} (n+1) u^{2n+3} = \frac{(n+1) h^2 a^{2n}}{r^{2n+3}}$$

Therefore, law of force $F \propto \frac{1}{r^{2n+3}}$, where $\mu = (n+1) h^2 a^{2n}$.



$$\begin{aligned}
 \text{Again } v^2 &= h^2 \left[u^2 + \left(\frac{du}{d\theta} \right)^2 \right] \\
 &= h^2 (u^2 + u^2 \tan^2 n\theta) \\
 &= h^2 u^2 \sec^2 n\theta = h^2 u^2 (a^n u^n)^2 \\
 &= h^2 a^{2n} u^{2n+2} = \frac{h^2 a^{2n}}{r^{2n+2}}
 \end{aligned}$$

$$\text{Law of velocity is } v = \frac{ha^n}{r^{n+1}} \Rightarrow v \propto \frac{1}{r^{n+1}}$$



Finally, the pedal equation is $\frac{h}{p} = \frac{ha^n}{r^{n+1}} \Rightarrow a^n p = r^{n+1}$.

Now, we see the cases where $n = 1, -1, 2, -2, -\frac{1}{2}, \frac{1}{2}, \dots$

Case I: $n = 1$

$$r = a \cos \theta \left[x^2 + y^2 = ax, \text{ a circle} \right]$$

So, law of force $F \propto \frac{1}{r^5}$, i.e., $F = \frac{\mu}{r^5}$, $\mu = 2h^2 a^2$

The velocity, $v = hau^2$, or $v \propto \frac{1}{r^2}$



The pedal equation, $ap = r^2$.

Case II: $n = -1$, then $\frac{1}{r} = \frac{1}{a} \cos \theta$ or $a = r \cos \theta$ or $x = a$

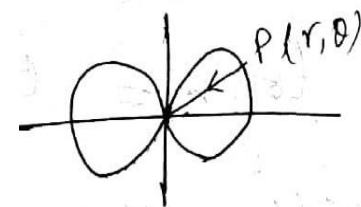
$$\mu = 0, F = 0$$

Case III: $n = 2 \Rightarrow r^2 = a^2 \cos 2\theta$ $\left[(x^2 + y^2)^2 = a^2 (x^2 - y^2), \text{ a Lemniscate} \right]$

$$F = 3h^2 a^4 u^7, \text{ or } F = \frac{\mu}{r^7} \Rightarrow F \propto \frac{1}{r^7}, \mu = 3h^2 a^4$$

$$\text{Velocity: } v = a^2 h u^3 \Rightarrow v \propto \frac{1}{r^3}$$

$$\text{Pedal equation: } a^2 p = r^3$$



Case IV: $n = -2$, a rectangular hyperbola.

$$\frac{1}{r^2} = \frac{1}{a^2} \cos 2\theta \Rightarrow a^2 = r^2 \cos 2\theta \quad [x^2 - y^2 = a^2]$$

$$F = -h^2 a^{-4} u^{-1} = -h^2 a^{-4} r \Rightarrow F \propto r, \mu = -h^2 a^{-4}$$

$$v = h^2 a^{-4} u^{-1} = h a^{-2} r \Rightarrow v \propto r$$

Pedal equation: $pr = a^2$.

$$\textbf{Case V: } n = -\frac{1}{2}, \frac{1}{\sqrt{r}} = \frac{\cos\left(\frac{\theta}{2}\right)}{\sqrt{a}} \Rightarrow \frac{2a}{r} = 1 + \cos \theta, \text{ a parabola.}$$

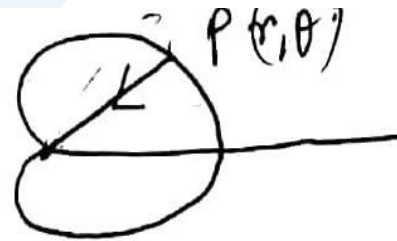


Case VI: $n = \frac{1}{2}$, $\sqrt{r} = \sqrt{a} \cos\left(\frac{\theta}{2}\right) \Rightarrow r = \frac{a}{2}(1 + \cos \theta)$, a cardioid .

$$F = \frac{3}{2}h^2au^4 \Rightarrow F \propto \frac{1}{r^4}, \mu = \frac{3}{2}h^2a$$

$$v = h\sqrt{au}^{3/2} \Rightarrow v \propto \frac{1}{r^{3/2}}$$

Pedal equation: $\sqrt{a}p = r^{3/2} \Rightarrow ap^2 = r^3$.



5) **Equianqular spiral:**

$$r = ae^{\theta \cot \alpha}, \quad a, \alpha \text{ being constants.}$$

$$\text{So, } \log r = \log a + \theta \cot \alpha$$

Differentiating w.r.t θ we get

$$\frac{1}{r} \frac{dr}{d\theta} = \cot \alpha \Rightarrow \cot \phi = \cot \alpha \Rightarrow \phi = \alpha \quad \left[\text{as } \tan \phi = r \frac{d\theta}{dr} \right]$$

$\therefore p = r \sin \phi = r \sin \alpha$, which is the pedal equation of the spiral.

The differential equation of central orbit in pedal form is $\frac{h^2}{p^3} \frac{dp}{dr} = F$.



$$\text{Therefore, } F = \frac{h^2}{p^3} \sin \alpha \left[\text{as } \frac{dp}{dr} = \sin \alpha \right]$$

$$= \frac{h^2 \sin \alpha}{r^3 \sin^3 \alpha} = \frac{h^2 \operatorname{cosec}^2 \alpha}{r^3}$$

$$\Rightarrow F \propto \frac{1}{r^3}, \text{ where } \mu = h^2 \operatorname{cosec}^2 \alpha.$$

$$v = \frac{h}{p} = \frac{h}{r \sin \alpha} = \frac{h \operatorname{cosec} \alpha}{r} \Rightarrow v \propto \frac{1}{r}.$$



Kepler's Laws of Motion:

Kepler, an astronomer (contemporary to Galileo) gave three laws of the motion of the planets round the sun.

The three laws are –

- (i) Each planet describes an elliptic orbit round the sun. Sun lies at one of its two foci.
- (ii) The area described by the radius vector drawn from the planet to the sun, in the same orbit, is proportional to the time of describing it.



- (iii) The square of the periodic time of the planet is proportional to the cube of the semi-major axis of its orbit.

Here the planet describes an elliptic central orbit with sun as centre of force. So, central force varies inversely to the square of the distance between the sun and the planet. Thus if F is the central force on the planet and r is the distance between the sun and the planet at any time t then

$$F = \frac{\mu}{r^2} \text{ with } \mu = \frac{h^2}{l} = \text{constant},$$



where l is the semi latus radius and h is the angular momentum per unit mass of the planet or twice of the sectorial area per unit time traced by the planet.

For the verification of third law, let a and b be the semi major and semi minor axis of the orbit. Then the area enclosed by the orbit is πab .

If T is the time period, then we have

$$T = \frac{\pi ab}{h/2} = \frac{2\pi ab}{h} = \frac{2\pi ab}{\sqrt{\mu l}} \frac{2\pi ab}{\sqrt{\mu b^2/a}}$$

$$\text{i.e., } T = \frac{2\pi}{\sqrt{\mu}} a^{3/2} \Rightarrow T^2 = \frac{4\pi^2}{\mu} a^3. \text{ So, } T^2 \propto a^3.$$



Validity of Third Law:

The third law is only true for any planet when the mass of the planet is negligible in comparison with that of the sun.

Proof: Let a planet of mass m revolves around the sun; let mass of the sun be M and G be the constant of gravitation.

The force of attraction between the two is $G \frac{m \times M}{r^2}$, r is the distance between the sun and the planet at any instance. Let g be the acceleration due to gravity of the planet, then $mg = G \frac{m \times M}{r^2}$.



So the acceleration of the planet is $g = G \frac{M}{r^2}$ towards the sun. Similarly, if

g' be that of the sun then $g' = G \frac{m}{r^2}$ towards the planet. Let us imposed an

acceleration g' on both the planet and the sun in the direction from the planet to the sun. This will make zero acceleration from the sun towards the planet and the resultant acceleration of the planet towards the sun will

become $(g + g') = G \frac{m + M}{r^2}$

Now it will be a central force under which the planet is orbiting the sun and so $\mu = (m + M)G$.



The angular momentum and the period will be

$$\sqrt{\mu l} \quad \text{and} \quad T = \frac{2\pi}{\sqrt{\mu}} a^{3/2}, \quad \text{i.e.,} \quad \sqrt{(m+M)Ga(1-e^2)} \quad \text{and} \quad \frac{2\pi}{\sqrt{(m+M)G}} a^{3/2}$$

respectively.

Now, let us consider two such planets. If T_1 and T_2 be time periods, m_1 and m_2 be the masses and a_1 and a_2 be the semi-major axes of the paths of the two planets, then

$$T_1 = \frac{2\pi}{\sqrt{(M+m_1)G}} a_1^{3/2}, T_2 = \frac{2\pi}{\sqrt{(M+m_2)G}} a_2^{3/2}$$



$$\text{Or, } \frac{M + m_1}{M + m_2} \frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3}$$

Now if Kepler's law is true, we have $\frac{M + m_1}{M + m_2} = 1$ which is possible only when m_1 and m_2 are negligible when compared to M , i.e. $M + m_1 \approx M$ and $M + m_2 \approx M$. Hence the result is proved.



Linear velocity and angular velocity at the ends of major axis and minor axis:

We know that in elliptic orbit $\frac{l}{r} = 1 + e \cos \theta$ the linear velocity at any position (r, θ) is given by $v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$ and the angular velocity is given by $\dot{\theta} = \frac{h}{r^2}$.

We also have $T = \frac{2\pi}{\sqrt{\mu}} a^{3/2}$ with $\mu = \frac{h^2}{l}$.

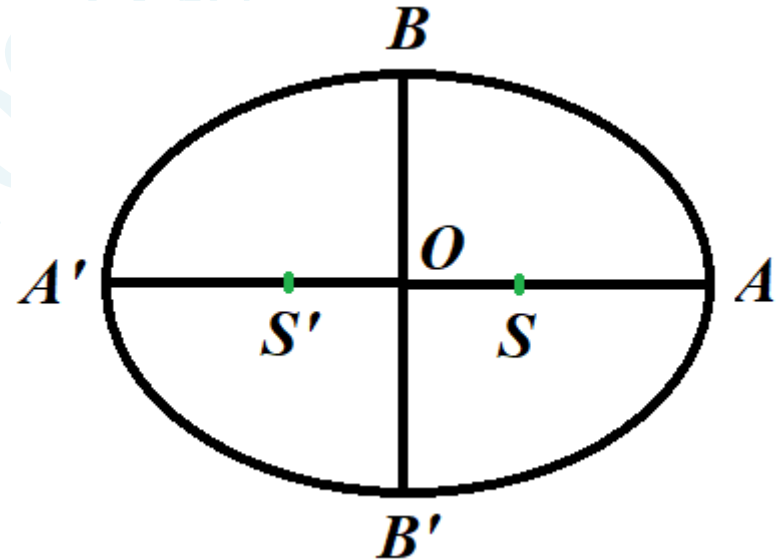


Now, let V_A and $V_{A'}$ be the velocities at the nearest and the farthest positions of a planet from the sun and those are the end points major axis of the elliptic orbit. Then the radius vectors at A and A' are $r_1 = a - ae$ and $r_2 = a + ae$.

Then we have

$$V_A^2 = \mu \left(\frac{2}{a - ae} - \frac{1}{a} \right) = \frac{\mu}{a} \frac{1+e}{1-e}$$

$$\text{and } V_{A'}^2 = \mu \left(\frac{2}{a + ae} - \frac{1}{a} \right) = \frac{\mu}{a} \frac{1-e}{1+e}$$



So, we have $V_A : V_{A'} = \frac{1+e}{1-e}$.

Similarly, when the planet is at one extremity of the minor axis B of its orbit, then the radius vector at B is given by

$$r_B^2 = b^2 + a^2 e^2 = a^2 \Rightarrow r_B = a \left[\because b^2 = a^2 - a^2 e^2 \right].$$

So, the linear velocity at this position will be

$$V_B^2 = \mu \left[\frac{2}{a} - \frac{1}{a} \right] = \frac{\mu}{a} = \frac{1}{2} \left(\frac{2\mu}{a} \right)$$

Thus velocity at B' , the other extremity of minor axis is the same.



Let the angular velocity at A and A' be ω_A and $\omega_{A'}$.

Then we have

$$\omega_A = \frac{h}{(a - ae)^2} = \frac{\sqrt{\mu l}}{a^2 (1 - e)^2} = \frac{\sqrt{\mu a (1 - e^2)}}{a^2 (1 - e)^2} = \frac{\sqrt{\mu} (1 + e)^{1/2}}{a^{3/2} (1 - e)^{3/2}} = \frac{2\pi (1 + e)^{1/2}}{T (1 - e)^{3/2}}$$

$$\omega_{A'} = \frac{h}{(a + ae)^2} = \frac{\sqrt{\mu l}}{a^2 (1 + e)^2} = \frac{2\pi (1 - e)^{1/2}}{T (1 + e)^{3/2}}$$

So, we have $\omega_A : \omega_{A'} = (1 + e)^2 : (1 - e)^2$.



Now, if ω_B and $\omega_{B'}$ is angular velocities at B and B' then we have

$$\omega_B = \omega_{B'} = \frac{\sqrt{\mu l}}{a^2} = \frac{\sqrt{\mu a(1-e^2)}}{a^2} = \frac{2\pi}{T} \sqrt{1-e^2}.$$

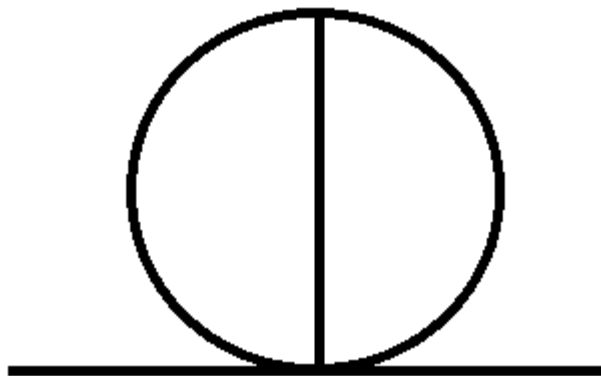


Part – VI

Motion on Smooth Vertical Curves



Circle

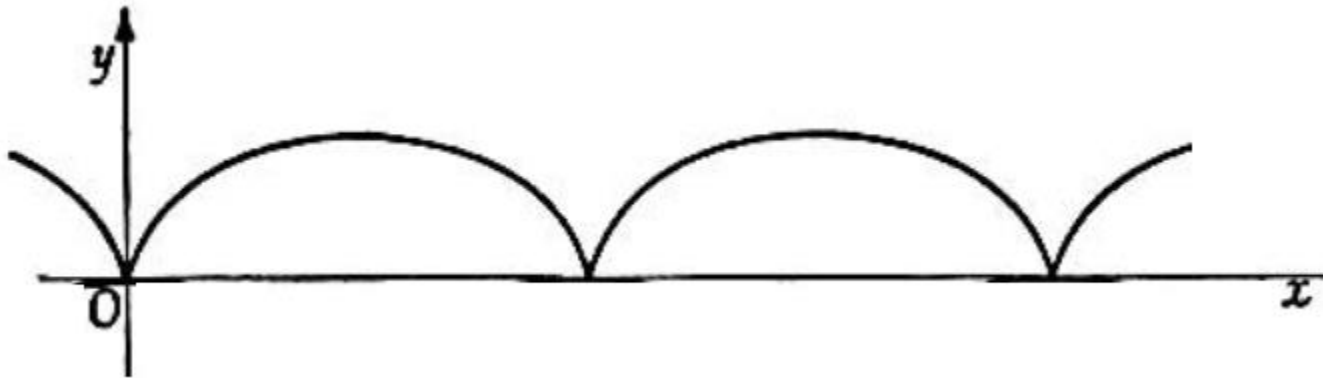


Parametric equation: $x = a \cos \theta, y = a \sin \theta;$

Intrinsic equation: $s = a\psi$

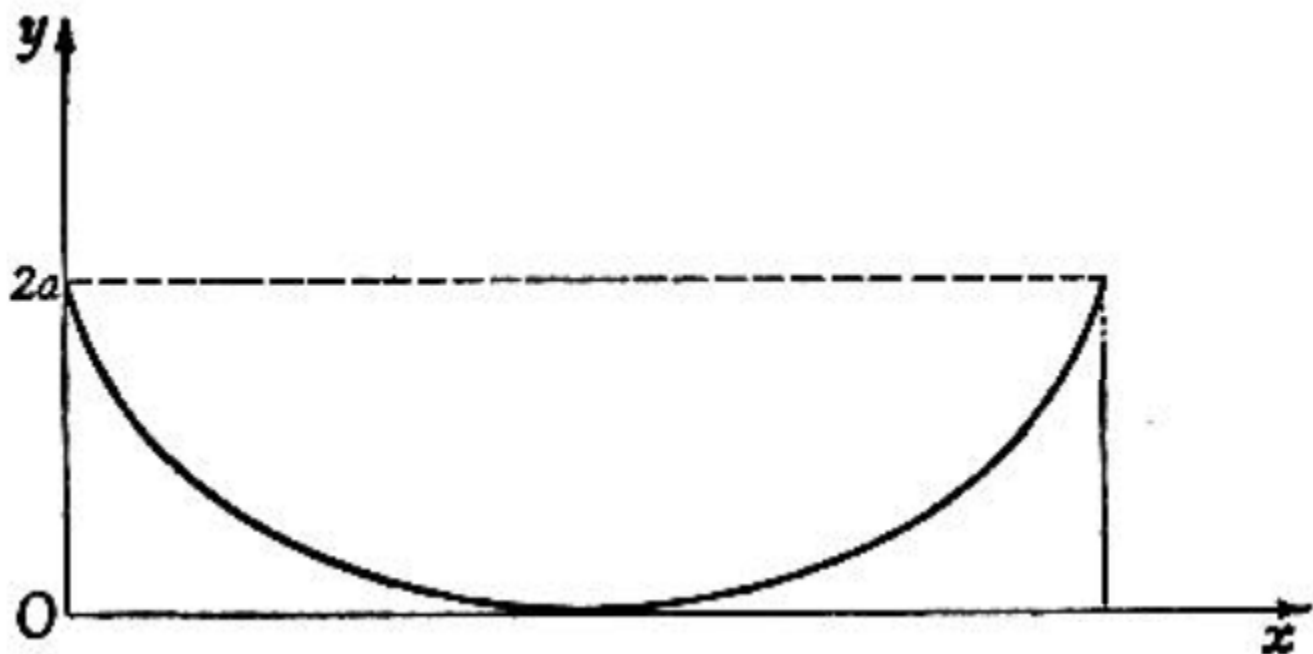


Cycloid



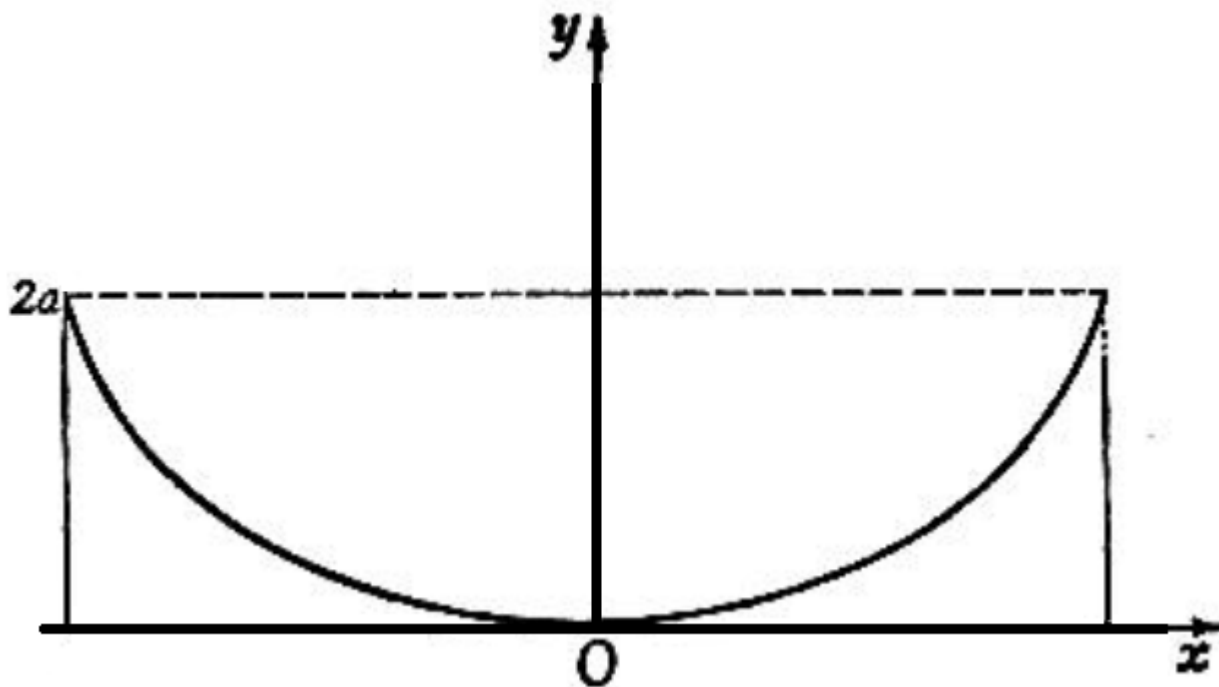
Parametric equation: $x = a(t - \sin t)$ and $y = a(1 - \cos t)$





Parametric equation: $x = a(t - \sin t)$ and $y = a(1 + \cos t)$





Parametric equation: $x = a(t + \sin t)$ and $y = a(1 - \cos t)$

Intrinsic equation: $s = 4a \sin \psi$; with $s^2 = 8ay$



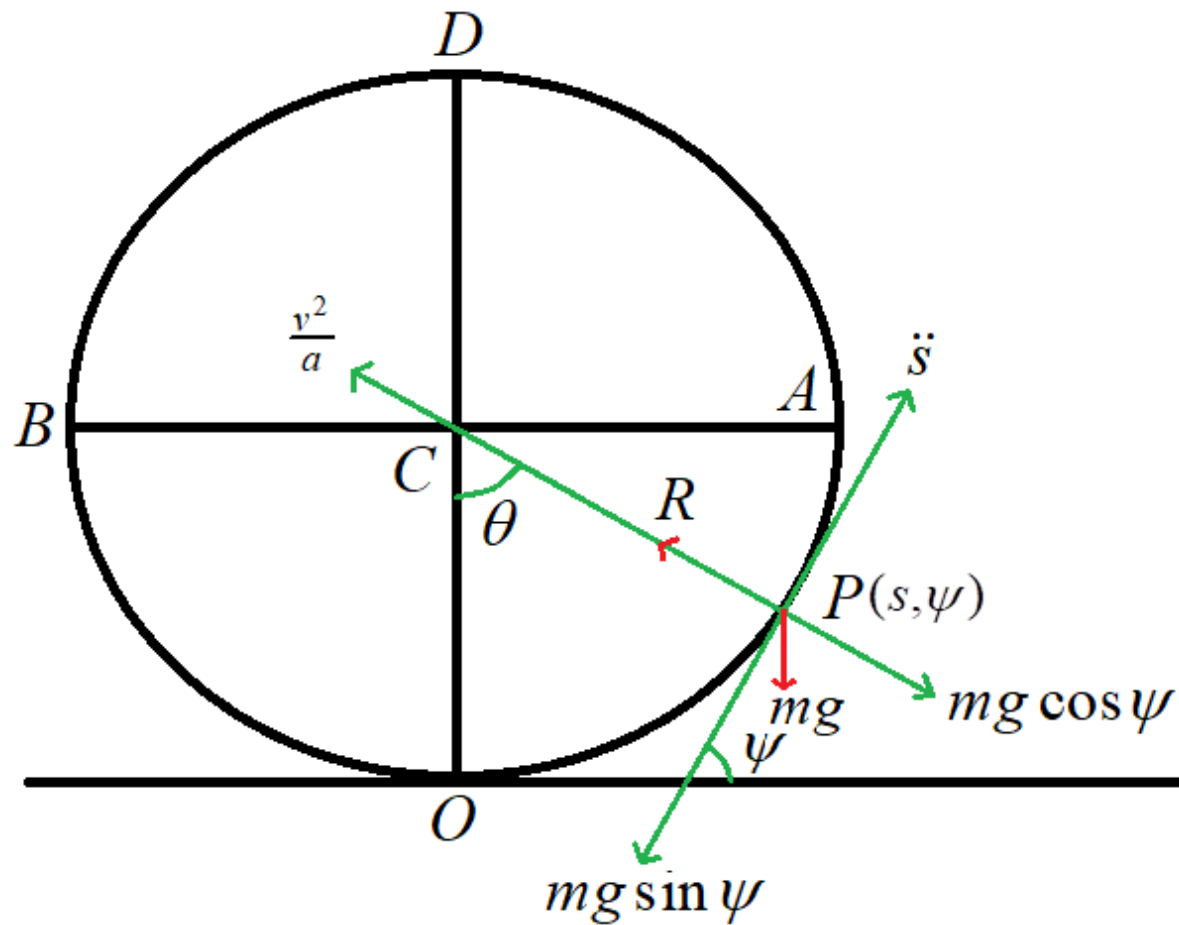
I) Vertical Motion on a Smooth Circle

A. Downward motion along the inner surface:

Let the motion occur towards the bottom point from one end of a horizontal diameter of a circle of radius a and under gravity only. The other force on the particle will be the reaction of the surface in the direction perpendicular to the tangent, i.e., along inward normal.

Let the particle of mass m be moving in the inner surface of the circle $OADB$ with the centre at C and let at time t $P(s, \psi)$ be the position of the particle. Then $OP=s$ and $s = a\psi$.





Then the equation of motion is

$$m\ddot{s} = -mg \sin \psi \quad \text{and} \quad m \frac{v^2}{a} = R - mg \cos \psi,$$

where R is the reaction.

So, we have $\ddot{s} = -g \sin \psi$ and $\frac{v^2}{a} = \frac{R}{m} - g \cos \psi$

If the radius CP makes an angle θ with the vertical diameter OD , then we have $\psi = \theta$ and $s = a\theta$. Then the equation becomes

$$\ddot{\theta} = -\frac{g}{a} \sin \theta \quad \dots\dots\dots (1) \quad \text{and} \quad \frac{v^2}{a} = \frac{R}{m} - g \cos \theta \quad \dots\dots\dots (2)$$



Integrating (1) we get

$$\dot{\theta}^2 = 2\frac{g}{a}\cos\theta + c_1 \Rightarrow (a\dot{\theta})^2 = 2ag\cos\theta + c_2 \Rightarrow v^2 = \dot{s}^2 = 2ag\cos\theta + c_2$$

Now, at $t = 0, \theta = \pi/2, v = 0 \Rightarrow c_2 = 0$

So, $v^2 = 2ag\cos\theta$ or, $(a\dot{\theta})^2 = 2ag\cos\theta$.

From (2),

$$2g\cos\theta = \frac{R}{m} - g\cos\theta \Rightarrow R = 3mg\cos\theta.$$

So, at the starting point the reaction is zero.



Now, if the time taken from $\theta = \frac{\pi}{2}$ to $\theta = 0$ is t_1 , then

$$\int_0^{t_1} dt = -\sqrt{\frac{a}{2g}} \int_{\theta=\frac{\pi}{2}}^{\theta=0} \sqrt{\sec \theta} d\theta$$

[$v = a\dot{\theta} = \pm\sqrt{2ag \cos \theta}$. For this case $a\dot{\theta} = -\sqrt{2ag \cos \theta}$, as t increases θ decreases, i.e., the motion along θ decreasing direction.]

Now, if the particle starts from a position very close to the vertex O , i.e., θ is very small, then we can write $\sin \theta \approx \theta$ then the equation (1) becomes

$$\ddot{\theta} = -\frac{g}{a} \theta$$



This represents a S.H.M about the bottom point with time period

$$\frac{2\pi}{\sqrt{\frac{g}{a}}}, \text{ i.e., } 2\pi\sqrt{\frac{a}{g}}.$$



B. Upward motion along the inner surface:

Let the particle be projected from the vertex O ($s = 0, \theta = 0$) with velocity u . The equations of motion are same as previous,

$$\text{i.e., } m\ddot{s} = -mg \sin \psi, \quad m \frac{v^2}{a} = R - mg \cos \psi$$

$$\text{i.e., } \ddot{\theta} = -\frac{g}{a} \sin \theta \quad \text{and} \quad m \frac{v^2}{a} = R - mg \cos \theta$$

Hence, we have $v^2 = 2ag \cos \theta + c$.



Now, at $\theta = 0, v = u \Rightarrow c = u^2 - 2ag$.

So, $v^2 = u^2 - 2ag(1 - \cos \theta)$. Therefore $R = m \left[\frac{u^2}{a} - 2g + 3g \cos \theta \right]$

Let at $\theta = \theta_1, u = 0$ and $\theta = \theta_2, R = 0$.

Now $v = 0 \Rightarrow \cos \theta_1 = 1 - \frac{u^2}{2ag}$ and $R = 0 \Rightarrow \cos \theta_2 = \frac{2}{3} \left(1 - \frac{u^2}{2ag} \right) = \frac{2}{3} \cos \theta_1$

Thus for $u^2 < 2ag, \cos \theta_1 > \cos \theta_2$ or $\theta_1 < \theta_2$.

These all are true up to $\theta = \pi/2$

So, $v = 0$ before $R = 0$ (up to $\theta = \pi/2$)



It means when particle comes at rest it remains in contact with the surface. Hence, it will start coming down. It will reach the lowest point with maximum velocity and will move on the other side up to same angle and again return back. Thus the particle starts oscillating about lowest point $s = 0$ or $\theta = 0$

Thus if the particle is projected with velocity less than square root of $2ag$, then it will come at rest at some position before A and the particle remains in contact with the surface.

In case $u^2 = 2ag, \cos \theta_1 = \cos \theta_2 = 0, \theta_1 = \theta_2 = \pi/2$



The particles come at rest and its contact with the surface at end A of horizontal diameter also vanishes. From this position it comes down and starts oscillating.

Now, when $u^2 > 2ag$. Also, let at the top the particle be at rest.

$$\text{Then } 0 = u^2 - 2ag(1 - \cos \pi) \Rightarrow u^2 = 4ag$$

Thus if the particles projected velocity is equal to $\sqrt{4ag}$ then it comes at rest at top position.

Then $R = m[2g + 3g \cos \theta]$ and it will be zero at

$$\theta = \cos^{-1}\left(-\frac{2}{3}\right) = +131.8^\circ.$$



It means for $u^2 = 4ag$ the contact of the particle with the surface vanishes before the top position.

Thus, the reaction become zero earlier to the linear velocity between $A(\theta = \pi/2)$ and $D(\theta = \pi)$ if $2ag < u^2 \leq 4ag$

If the reaction at the top is zero, then

$$0 = m \left[\frac{u^2}{a} - 2g + 3g \cos \pi \right] \Rightarrow u^2 = 5ag$$

It means if the particle reaches the top then it's projected velocity is equal to $\sqrt{5ag}$.



From it one may conclude that if $u^2 > 5ag$ then particle will move on the other side from the top position and will then complete one round.

Then the reaction of the bottom position is given by

$$R > m \left[\frac{5ag}{a} - 2g + 3g \right] \text{ or } R > 6amg$$

Further if $u^2 > 5ag$, let $u^2 = 6ag$ then

$$v^2 = 4ag + 2ag \cos \theta$$

$$R = \frac{m}{a} (4ag + 3ag \cos \theta)$$



So, at $\theta = \frac{\pi}{3}$, $v^2 = 5ag$ and $R = 5.5mg$

$\theta = \frac{\pi}{2}$, $v^2 = 4ag$ and $R = 4mg$

$\theta = \frac{2\pi}{3}$, $v^2 = 3ag$ and $R = 2.5mg$

$\theta = \pi$, $v^2 = 2ag$ and $R = mg$.

Thus one may observed that the velocity and the reaction on the particle, both decrease up to $\theta = \pi$. Beyond that both again increase and at the lowest are

$$v^2 = 4ag + 2ag \cos 2\pi = 6ag = u^2 \quad R = \frac{m}{a} [4ag + 3ag \cos 2\pi] = 7mg.$$



***Time of coming at position of no contact when projected with $u^2 = 4ag$**

If θ_1 be angle at the point of no contact, then

$$m \left[\frac{4ag}{a} - 2g + 3g \cos \theta_1 \right] = 0 \Rightarrow \cos \theta_1 = -\frac{2}{3}. \text{ So, } \theta_1 = \cos^{-1} \left(-\frac{2}{3} \right).$$

We have $v^2 = 4ag - 2ag(1 - \cos \theta)$

$$\Rightarrow (a\dot{\theta})^2 = 2ag(1 + \cos \theta) = 4ag \cos^2 \frac{\theta}{2}$$

$$\Rightarrow a\dot{\theta} = +2\sqrt{ag} \cos \frac{\theta}{2} \text{ [as } t \text{ increases } \theta \text{ decrease]}$$

$$\Rightarrow \frac{\dot{\theta}}{\cos \frac{\theta}{2}} = \sqrt{\frac{4g}{a}}$$



If T be the required time, then

$$T\sqrt{\frac{4g}{a}} = \int_0^{\theta_1} \frac{d\theta}{\cos\left(\frac{\theta}{2}\right)} = 2 \left[\log \left(\sec \frac{\theta}{2} + \tan \frac{\theta}{2} \right) \right]_0^{\theta_1}$$

$$\therefore T = \sqrt{\frac{a}{g}} \log \left(\sec \frac{\theta_1}{2} + \tan \frac{\theta_1}{2} \right)$$

$$\text{i.e., } T = \sqrt{\frac{a}{g}} \log \left(\sqrt{6} + \sqrt{5} \right).$$

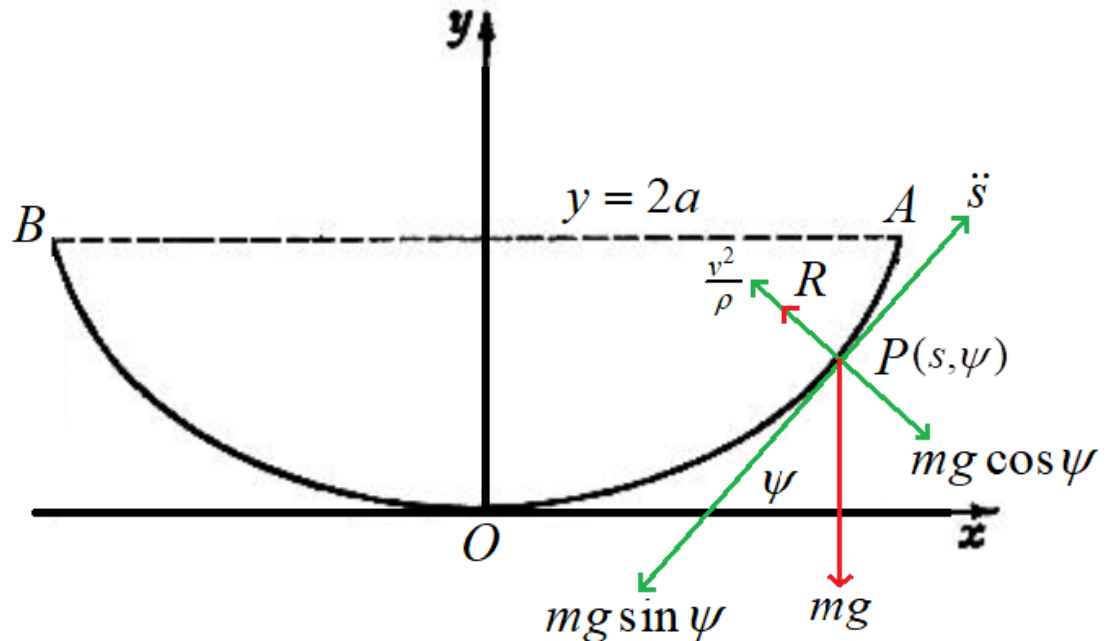


II) Motion on a vertical smooth cycloid

A. Downward motion along the inner surface:

Let a particle be moved along the inner surface in a direction from cusp $A(s = 4a)$ towards the vertex $O(s = 0)$ of the cycloid $s = 4a \sin \psi$.

Then the equations of motion are



$$m\ddot{s} = -mg \sin \psi \quad \text{and} \quad m \left(\frac{v^2}{\rho} \right) = R - mg \cos \psi$$

$$\Rightarrow \ddot{s} = -g \sin \psi \quad \dots\dots\dots (1) \quad \text{and} \quad \frac{v^2}{\rho} = \frac{R}{m} - g \cos \psi \quad \dots\dots\dots (2)$$

We have from (1) and the equation of cycloid

$$\ddot{s} = -\frac{g}{4a} s,$$

which shows that the motion of the particle is SHM about the vertex O of

$$\text{the time period } T = 2\pi \sqrt{\frac{4a}{g}} = 4\pi \sqrt{\frac{a}{g}}.$$



Its general solution may be written as $s = c_1 \cos \sqrt{\frac{g}{4a}}t + c_2 \sin \sqrt{\frac{g}{4a}}t$.

Also we have $v^2 = \dot{s}^2 = -\frac{g}{4a}s^2 + c_3$.

Using $t = 0, s = 4a$ and $t = 0, \dot{s} = 0$, we get $c_2 = 0, c_1 = 4a, c_3 = +4ag$

Thus $s = 4a \cos \sqrt{\frac{g}{4a}}t$ and $\dot{s} = v = -2\sqrt{ag} \sin \sqrt{\frac{g}{4a}}t$.

Also, $\dot{s}^2 = v^2 = \frac{g}{4a}(16a^2 - s^2) = \frac{g}{4a}(16a^2 - 16a^2 \sin^2 \psi)$

i.e., $v^2 = 4ag \cos^2 \psi \Rightarrow v = -2\sqrt{ag} \cos \psi$

$$\left[v_{\max} = 2\sqrt{ag} \text{ at vertex } s = 0, \psi = 0 \right]$$



$$\begin{aligned}
 \text{and } R &= m \left[\frac{v^2}{\rho} + g \cos \psi \right] \\
 &= m \left[\frac{4ag \cos^2 \psi}{4a \cos \psi} + g \cos \psi \right] \quad \left[\text{as } \rho = \frac{ds}{d\psi} = 4a \cos \psi \right] \\
 &= m [g \cos \psi + g \cos \psi]
 \end{aligned}$$

i.e., $R = 2mg \cos \psi$

This the reaction at $P(s, \psi)$.



***Time period for describing different arc lengths**

At first the time in reaching the vertex.

Let it be T_1 . Then

$$0 = 4a \cos \sqrt{\frac{g}{4a}} T_1 \Rightarrow T_1 = \frac{\pi}{2} \sqrt{\frac{4a}{g}} = \pi \sqrt{\frac{a}{g}} = \frac{T}{4}.$$

Let the time required in coming down to a position $s = 2a$, $\psi = \frac{\pi}{3}$, i.e., half of the arc length OA be T_2 . Then we have

$$2a = 4a \cos \sqrt{\frac{g}{4a}} T_2 \Rightarrow T_2 = \frac{\pi}{3} \sqrt{\frac{4a}{g}} = \frac{2\pi}{3} \sqrt{\frac{a}{g}}.$$



Again, let the time required in coming to the position whose vertical height is a be T_3 . Then we have $s^2 = 8ay$. So, $s^2 = 8a^2 \Rightarrow s = 2\sqrt{2}a$

$$\text{and } 2\sqrt{2}a = 4a \sin \psi \Rightarrow \psi = \frac{\pi}{4}$$

$$\text{Hence, } 2\sqrt{2}a = 4a \cos \sqrt{\frac{g}{4a}} T_3 \Rightarrow T_3 = \frac{\pi}{4} \sqrt{\frac{4a}{g}} = \frac{\pi}{2} \sqrt{\frac{a}{g}}.$$

If T_4 be time taken by the particle in moving from $s = 2\sqrt{2}a$ to $s=0$, then

$$T_4 = T_1 - T_3 = \frac{\pi}{2} \sqrt{\frac{a}{g}}.$$

$$\text{So, } T_3 = T_4$$



***Angular Velocity at a point, $\dot{\psi}$**

We have $\dot{s} = \frac{ds}{dt} = \frac{ds}{d\psi} \frac{d\psi}{dt} = \rho \dot{\psi}$

So, $\dot{\psi} = \frac{\dot{s}}{\rho} = \frac{v}{\rho} = \frac{\sqrt{4ag \cos \psi}}{4a \cos \psi} = \sqrt{\frac{g}{4a}}$.

***Change initial velocity**

Let the particle start from $s = 4a$ with velocity u , then the initial condition will be $t = 0, s = 4a$ and $t = 0, \dot{s} = -u$, where negative sign is assigned as the motion occurs in s decreasing.



Then we have $c_1 = 4a$ and $c_2 = -u\sqrt{\frac{4a}{g}}$

Thus, $s = 4a \cos \sqrt{\frac{g}{4a}}t - u\sqrt{\frac{4a}{g}} \sin \sqrt{\frac{g}{4a}}t$

If the time of reaching the vertex is T_5 , then

$$0 = 4a \cos \left(\sqrt{\frac{g}{4a}}T_5 \right) - u\sqrt{\frac{4a}{g}} \sin \left(\sqrt{\frac{g}{4a}}T_5 \right)$$

$$\Rightarrow \tan \left(\sqrt{\frac{g}{4a}}T_5 \right) = \frac{4a}{u} \sqrt{\frac{g}{4a}} = \frac{\sqrt{4ag}}{u} \Rightarrow T_5 = \sqrt{\frac{4a}{g}} \tan^{-1} \left(\frac{\sqrt{4ag}}{u} \right).$$



***Another initial condition**

Let the particle be started from rest position from $s = b < 4a$

Then we have $s = b \cos \sqrt{\frac{g}{4a}}t$ and $\dot{s} = -b \sqrt{\frac{g}{4a}} \sin \sqrt{\frac{g}{4a}}t$.

***Simultaneous motion of two particles**

Let two particles start motion from the cusp from rest position at the interval of time t^* . [Then to find the time when both will meet]



If s_1 be the arc travelled by the first particle after time t then

$$s_1 = 4a \cos \sqrt{\frac{g}{4a}} t$$

and s_2 be that by the second particle then

$$s_2 = 4a \cos \sqrt{\frac{g}{4a}} (t - t^*).$$

At a position where both the particle meet, we have $s_1 = s_2$

$$\text{i.e., } 4a \cos \sqrt{\frac{g}{4a}} t = 4a \cos \sqrt{\frac{g}{4a}} (t - t^*) \Rightarrow \sqrt{\frac{g}{4a}} t = 2\pi - \sqrt{\frac{g}{4a}} (t - t^*)$$



$$\Rightarrow \sqrt{\frac{g}{a}}t = 2\pi + \sqrt{\frac{g}{4a}}t^* \Rightarrow t = 2\pi\sqrt{\frac{a}{g}} + \frac{1}{2}t^*$$

$$\Rightarrow t = \frac{T}{2} + \frac{t^*}{2} = \frac{1}{2}(T + t^*)$$



B. Upward motion along the inner surface:

Let the particle be projected from the vertex ($s = 0$) with velocity u along the inner surface towards the cusp $A(s = 4a)$

Let at time t $P(s, \psi)$ be position of the particle. Then

$$m\ddot{s} = -mg \sin \psi \quad \text{and} \quad m \frac{v^2}{\rho} = R - mg \cos \psi$$

For the cycloid we have

$$\ddot{s} = -\frac{g}{4a} s \quad \left[\text{as } s = 4a \sin \psi \right]$$



$$\dot{s}^2 = v^2 = -\frac{g}{4a}s^2 + c_4 \Rightarrow v^2 = u^2 - \frac{g}{4a}s^2 \text{ [using } \dot{s} = u \text{ at } s = 0 \text{]}$$

So, the motion of the particle will be a SHM with period $4\pi\sqrt{\frac{a}{g}}$ and amplitude S . The arc length S will depend on u .

If the particle reaches up to cusp and comes at rest there, then $S = 4a, v = 0$. Then $u^2 = 4ag$.

And if the particle reaches up to the position whose vertical height is a , then $s = 2\sqrt{2}a, v = 0$ (and $S = 2\sqrt{2}a$), which implies $u^2 = \frac{g}{4a}8a^2 = 2ag$.



The expression of R at a point is

$$R = mg \cos \psi + m \frac{1}{\rho} \left(u^2 - \frac{g}{4a} s^2 \right) = mg \cos \psi + \frac{m}{4a \cos \psi} \left(u^2 - \frac{g}{4a} s^2 \right).$$

So, for $u^2 = 4ag$, the reaction at cusp, $\psi = \frac{\pi}{2}$ is

$$R = mg \cos \frac{\pi}{2} + \frac{m}{4a} (4ag - 4ag) = 0$$

and at $\psi = \frac{\pi}{4}$ ($s = 2\sqrt{2}a$ or $y = a$)



$$R = mg \cos \frac{\pi}{4} + \frac{m}{4a \cos \frac{\pi}{4}} \left(4ag - \frac{g}{4a} 8a^2 \right)$$

$$\Rightarrow R = \frac{mg}{\sqrt{2}} + \frac{m}{4a \frac{1}{\sqrt{2}}} (4ag - 2ag) = \frac{mg}{\sqrt{2}} + \frac{m}{\sqrt{2}} = \sqrt{2}mg.$$

i.e., the particle is in contact with the surface in all the positions below the cusp.



***Time taken in describing certain arc length**

We have the general solution as $s = c_1 \cos \sqrt{\frac{g}{4a}}t + c_2 \sin \sqrt{\frac{g}{4a}}t$

Using $t = 0, s = 0, \dot{s} = u$, we have $c_1 = 0$ and $c_2 = u \sqrt{\frac{4a}{g}}$

Thus $s = u \sqrt{\frac{4a}{g}} \sin \sqrt{\frac{g}{4a}}t$. Now, if T_1 be the time required to reach the cusp

when the particle is projected with $u = \sqrt{4ag}$, then we have

$$4a = \sqrt{4ag} \sqrt{\frac{4a}{g}} \sin \sqrt{\frac{g}{4a}}T_1 \Rightarrow 1 = \sin \sqrt{\frac{g}{4a}}T_1 \Rightarrow T_1 = \frac{\pi}{2} \sqrt{\frac{4a}{g}} = \pi \sqrt{\frac{a}{g}} = \frac{T}{4}.$$



***To get the intrinsic equation of cycloid and to prove $s^2 = 8ay$**

We have $x = a(t + \sin t)$ and $y = a(1 - \cos t)$.

$$\text{So, } \frac{dx}{dt} = a(1 + \cos t), \frac{dy}{dt} = a \sin t$$

$$\text{and } \frac{dy}{dx} = \frac{\sin t}{1 + \cos t} \Rightarrow \tan \psi = \frac{2 \sin \frac{t}{2} \cos \frac{t}{2}}{2 \cos^2 \frac{t}{2}} \left[\frac{dy}{dx} = \tan \psi \right]$$

$$\Rightarrow \tan \psi = \tan \frac{t}{2} \Rightarrow \psi = \frac{t}{2}$$

$$ds = \sqrt{dx^2 + dy^2} \Rightarrow ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$



$$\Rightarrow ds = a\sqrt{(1 + \cos t)^2 + \sin^2 t} dt = a\sqrt{2 + 2\cos t} dt$$

$$\Rightarrow ds = 2a \cos \frac{t}{2} dt \Rightarrow ds = (2a \cos \psi)(2d\psi) \quad \left[\text{as } \psi = \frac{t}{2}, d\psi = \frac{dt}{2} \right]$$

$$\Rightarrow ds = 4a \cos \psi d\psi$$

$\Rightarrow s = 4a \sin \psi$, the intrinsic equation of the cycloid.

$$\text{Now, } y = a(1 - \cos t) = a2\sin^2 \frac{t}{2} = 2a \sin^2 \psi$$

$$\Rightarrow y = 2a \left(\frac{s}{4a} \right)^2 \Rightarrow y = \frac{s^2}{8a} \Rightarrow s^2 = 8ay \quad (\text{Proved})$$



Part – VII

Motion in a Moving Frame of Reference



A moving cycle or a running car or a running train and their rotating wheels, a flying bird, a rotating disc, these all are examples of moving frames for an observer in a fixed frame. But for a passenger sitting or moving in a compartment of a running train, the compartment is not a moving frame, instead it is a fixed frame.

We have to see whether the Newton's Laws of motion are applicable on the object at rest or changing its position in a moving frame or not and whether there is any additional force or not on the object.

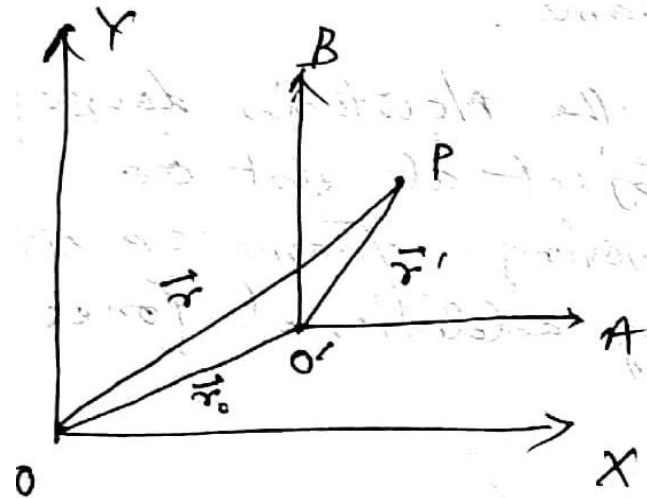


Motion in a Linearly Moving Frame

A) Linear motion with uniform velocity:

Let us consider an object of mass m moving in a plane $O'AB$. This frame itself is moving with uniform linear velocity with reference to a fixed plane of reference OXY .

Let any time t , the coordinates of the object at position P , with reference to fixed plane and moving plane be $\vec{r}(x, y)$ and $\vec{r}'(x', y')$ respectively, where as those of O' with



reference to O be $\vec{r}_o(x_o, y_o)$. Then

$$\vec{r} = \vec{r}' + \vec{r}_o, \text{ or } x = x' + x_o, y = y' + y_o.$$

Differentiating the above expressions w.r.t time variable w.r.t the fixed frame of reference we have

$$\vec{v} = \vec{v}' + \vec{v}_o, \text{ or } \dot{x} = \dot{x}' + \dot{x}_o, \dot{y} = \dot{y}' + \dot{y}_o, \quad (\text{i})$$

where the terms from the left to right represent velocities of the particle with reference to fixed plane OXY moving plane $O'AB$ and uniform velocity of the moving plane with reference to the fixed plane, respectively.



Further differentiating w.r.t time variable (in the fixed frame of reference) we have the two accelerations equal to each other, i.e., $\vec{a} = \vec{a}'$ as $\dot{\vec{v}}_o = 0$.

Thus according to Newton's law, the equation of motion of the object w.r.t the fixed plane $m\vec{a} = \vec{F}$ may also be written as $m\vec{a}' = \vec{F}$, where $\vec{F}$ is external force acting on the particle.

It means all the frames moving with uniform linear velocity relative to a fixed frame are also a Newtonian frame of references (in which Newton's Laws of motion hold good). So, there is no effect of the uniform linear motion of the frame on the motion of the object in it.



B) Linear Motion with an acceleration:

Let the plane $O'AB$ be moving with an acceleration $\vec{a}$ with reference to the fixed plane. Then differentiating (i) w.r.t time (w.r.t fixed plane), we have

$$\vec{a} = \vec{a}' + \vec{a}_o, \text{ or } \ddot{x} = \ddot{x}' + \ddot{x}_o, \ddot{y} = \ddot{y}' + \ddot{y}_o,$$

where first two terms from left to right represent the accelerations of the particle with reference to fixed plane and moving plane, respectively. So, the equation of motion of the object w.r.t the fixed plane is

$$m\vec{a} = \vec{F}, \text{ i.e., } m\vec{a}' = \vec{F} - m\vec{a}_o.$$



Since the term on the left hand side represents the product of mass and acceleration (effective force) of the object with reference to moving plane, therefore the equation represent the equation of motion of the object with reference to $O'AB$. Comparing with the previous case, we observe that an additional force $(-m\vec{a}_o)$ comes into effect. Obviously, it is due to the linear motion of $O'AB$ with an acceleration.

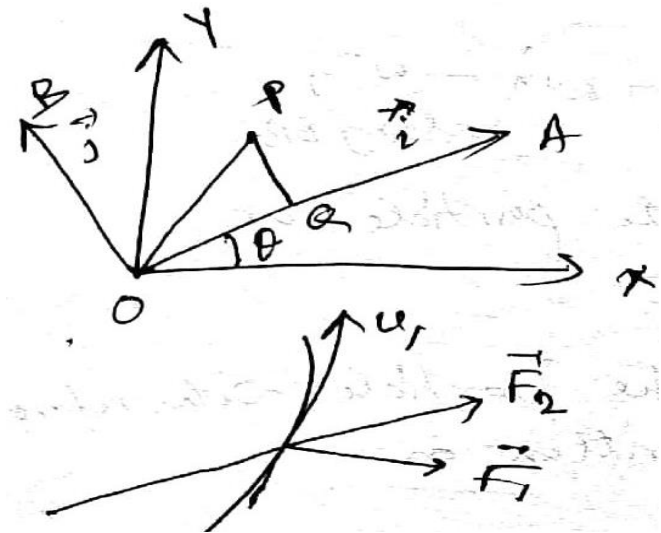
So we may conclude that a frame moving linearly with acceleration is not a Newtonian Frame of Reference. The force is a fictitious force and it vanishes when the acceleration vanishes.



Motion in a Uniformly Rotating Plane

Let a two dimensional frame of reference $O'AB$ be rotating with uniform angular velocity $\vec{\omega}$ in its plane. Let its axis of rotation pass through its origin. Let it initially be coincident with the fixed frame OXY , origin of the both frames being coincident. So, we can name the rotating frame as OAB .

Let $P(\vec{r})$ be the position vector of an object moving in the rotating frame at any time t . Further, let its components in rotating plane



OAB be $x=OQ$ and $y=PQ$.

Let any time the axis OA make angle θ with the fixed axis OA .

Let $\vec{i}$ and $\vec{j}$ be unit vectors along two axes OA and OB respectively.

Then we have $\vec{r} = x\vec{i} + y\vec{j}$.

The rate of change of $\vec{r}$ with time reference to the fixed frame OXY will be

$$\frac{d\vec{r}}{dt} = \dot{\vec{r}} = \dot{x}\vec{i} + \dot{y}\vec{j} + x\frac{d\vec{i}}{dt} + y\frac{d\vec{j}}{dt}.$$

We know that $\frac{d\vec{i}}{dt} = \frac{d\vec{i}}{d\theta} \frac{d\theta}{dt} = \omega\vec{j}$ and $\frac{d\vec{j}}{dt} = \frac{d\vec{j}}{d\theta} \frac{d\theta}{dt} = -\omega\vec{i}$,



where ω is the magnitude of the angular velocity vector along the axis of rotation and is the rate of change with respect to time variable w.r.t the fixed frame of the angular displacement of the rotating frame.

$$\text{So, } \dot{\vec{r}} = (\dot{x} - \omega y)\vec{i} + (\dot{y} + \omega x)\vec{j},$$

i.e., the velocity components of the object at P are $(\dot{x} - \omega y)$ and $(\dot{y} + \omega x)$ along OA and OB respectively, when observed from the fixed plane.

Again differentiating w.r.t. t , the acceleration of the object with reference to the fixed frame will have the expression

$$\ddot{\vec{r}} = (\ddot{x} - \omega \dot{y})\vec{i} + (\ddot{y} + \omega \dot{x})\vec{j} + (\dot{x} - \omega y)\omega\vec{j} + (\dot{y} + \omega x)(-\omega\vec{i}),$$



$$\text{i.e., } \ddot{\vec{r}} = (\ddot{x} - 2\omega\dot{y} - \omega^2 x)\vec{i} + (\ddot{y} + 2\omega\dot{x} - \omega^2 y)\vec{j}.$$

It means the acceleration components along the rotating axes are

$$\ddot{x} - 2\omega\dot{y} - \omega^2 x \text{ along } OA \text{ and } \ddot{y} + 2\omega\dot{x} - \omega^2 y \text{ along } OB.$$

Let the external force acting on the particle be $\vec{F} = X\vec{i} + Y\vec{j}$

Then the equation of motion of the particle with reference to fixed frame OXY may be written as

$$m\ddot{\vec{r}} = \vec{F}$$

$$\text{Or, } m(\ddot{x} - 2\omega\dot{y} - \omega^2 x) = X \text{ and } m(\ddot{y} + 2\omega\dot{x} - \omega^2 y) = Y$$

$$\text{Or, } m\ddot{x} = X + 2m\omega\dot{y} + m\omega^2 x \text{ and } m\ddot{y} = Y - 2m\omega\dot{x} + m\omega^2 y$$



If OAB plane is at rest and coincides with OXY the above equation of motion reduces to those in accordance with Newton's laws of motion, i.e., reduce to

$$m\ddot{x} = X \text{ and } m\ddot{y} = Y.$$

But, the plane OAB is rotating w.r.t fixed plane OXY , hence the equations of motion of the object in the rotating plane w.r.t fixed frame may be written as

$$m\ddot{x} = X + X_1 + X_2 \text{ and } m\ddot{y} = Y + Y_1 + Y_2,$$

where $F_1(X_1, Y_1)$ and $F_2(X_2, Y_2)$ may be interpreted as two additional fictitious forces introduced due to the rotating of the plane.



Thus the plane rotating about its origin, even with uniform angular velocity in its own plane is not a Newtonian frame of reference.

The force $\vec{F}_1$ is of magnitude $2m\omega v_1$, where $\vec{v}_1 = \dot{x}\vec{i} + \dot{y}\vec{j}$ is the linear velocity of the particle along the tangent to the path described by the particle in the rotating plane. It is known as Coriolis force and is perpendicular to direction of $\vec{v}_1$ in a sense opposite to that of angular velocity. Also the second force $\vec{F}_2$ has magnitude $m\omega^2 r$ and acts along the radial direction. It is referred to as Centrifugal force.



If the object is at rest in the rotating frame it will experience only centrifugal force.

